

Knowledge and Reasoning

Knowledge and Reasoning → humans are very good at acquiring new information by combining raw knowledge, experience, with reasoning;

Examples:

- Diagnosis –e.g., a physician diagnoses a patient, i.e., it infers what disease he/she has, based on the knowledge he/she acquired as a student, textbooks, prior cases and also some reasoning process (patterns of association, or other process) that he/she may not be able to describe.
- Car repair diagnosis
- Common sense reasoning
- Inventions, new ideas

Key issues:

- 1- Representation of knowledge
- 2- Reasoning and inference processes

Knowledge-base Agents

Representation of knowledge → knowledge base

Knowledge base = set of sentences in a formal language
representing facts about the world(*)

(*) called knowledge representation language

Knowledge-base Agents

Key issues:

- Representation of knowledge → knowledge base
- Reasoning processes → inference/reasoning

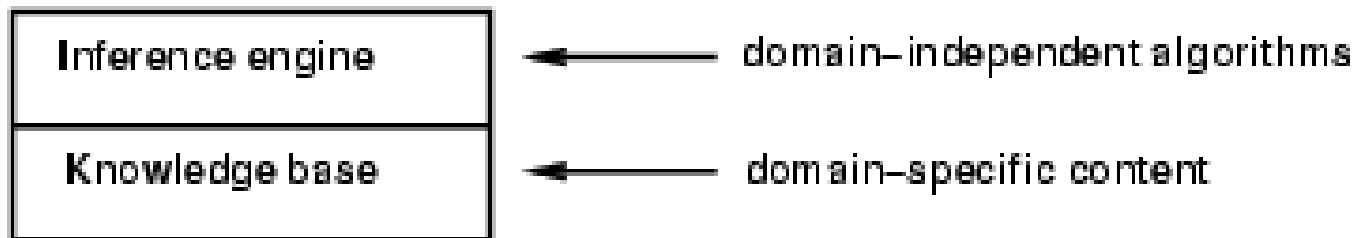
Knowledge base = set of sentences in a formal language representing facts about the world(*)

(*) called knowledge representation language

Knowledge bases

Reasoning processes → **inference/reasoning** mechanisms to query what is known, to derive new information, to make decisions →
i.e. how to derive new sentences from old sentences

Logical agents – **inference** must obey the fundamental requirement that when one asks a question to the knowledge base, the **answer should follow from what has been told to the knowledge base previously**. (In other words the inference process should not “make things” up...)



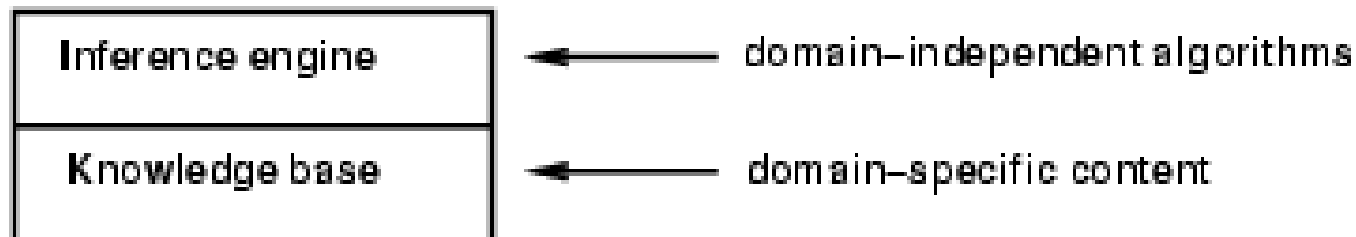
Knowledge bases

Key aspects:

- How to add sentences to the knowledge base
- How to query the knowledge base

Both tasks may involve **inference** – i.e. how to derive new sentences from old sentences

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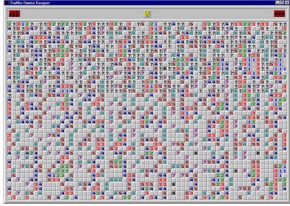
A simple knowledge-based agent

```
function KB-AGENT(percept) returns an action  
  static: KB, a knowledge base  
           t, a counter, initially 0, indicating time  
  
  TELL(KB, MAKE-PERCEPT-SENTENCE(percept, t))  
  action ← ASK(KB, MAKE-ACTION-QUERY(t))  
  TELL(KB, MAKE-ACTION-SENTENCE(action, t))  
  t ← t + 1  
  return action
```

The agent must be able to:

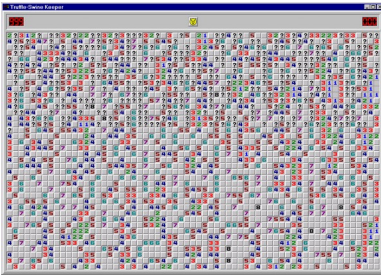
- Represent states, actions, etc.
- Incorporate new percepts
- Update internal representations of the world
- Deduce hidden properties of the world
- Deduce appropriate actions

—

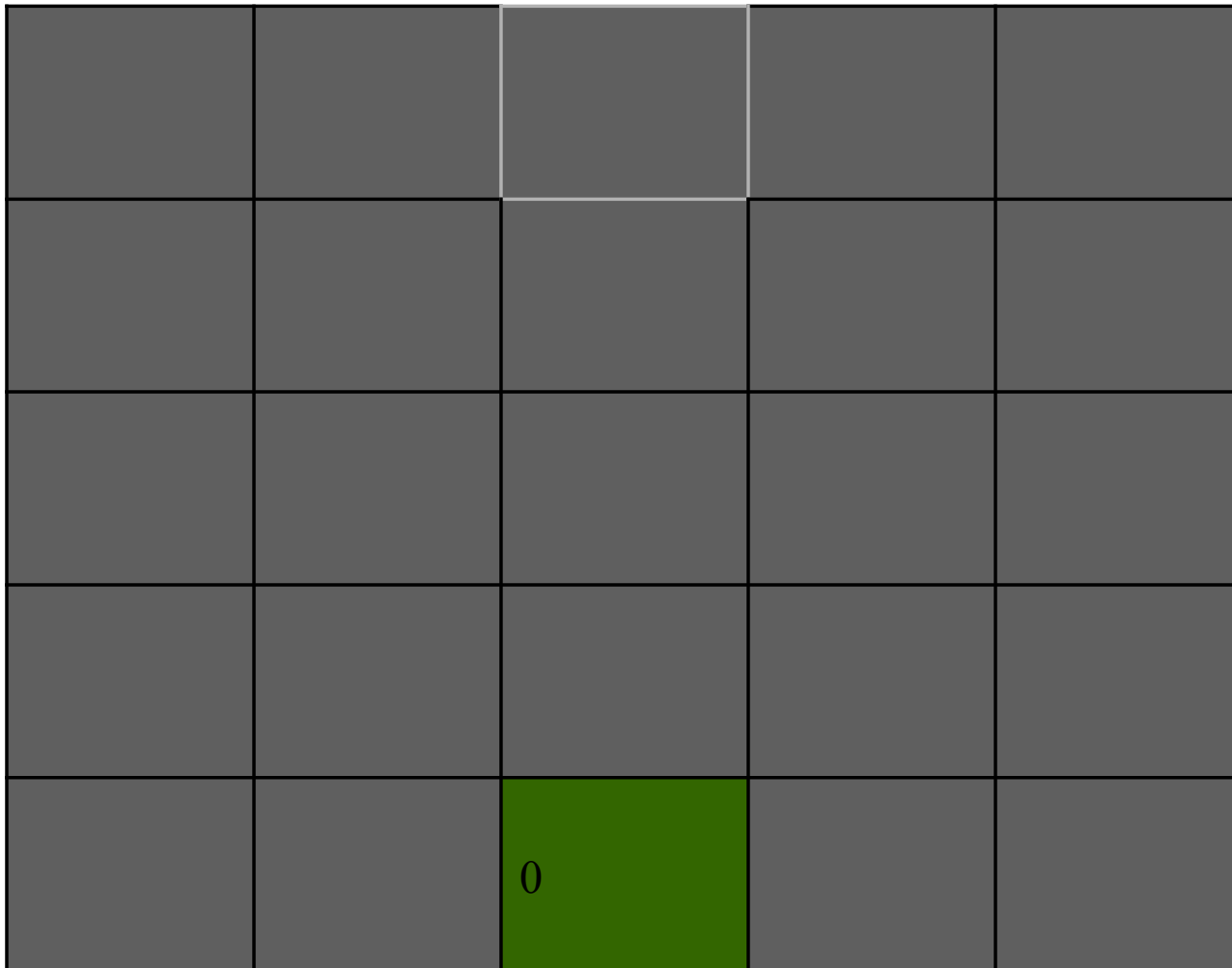


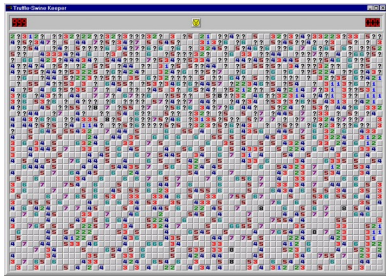
Mine Sweeper

You have to find a way from a start field to an exit field on grid where mines are hidden. An electronic device tells you how many mines are hidden on the neighboring fields (vertically, horizontally, and diagonally). However, the device is unable to give the exact location of the mines that it has detected. How do you proceed?

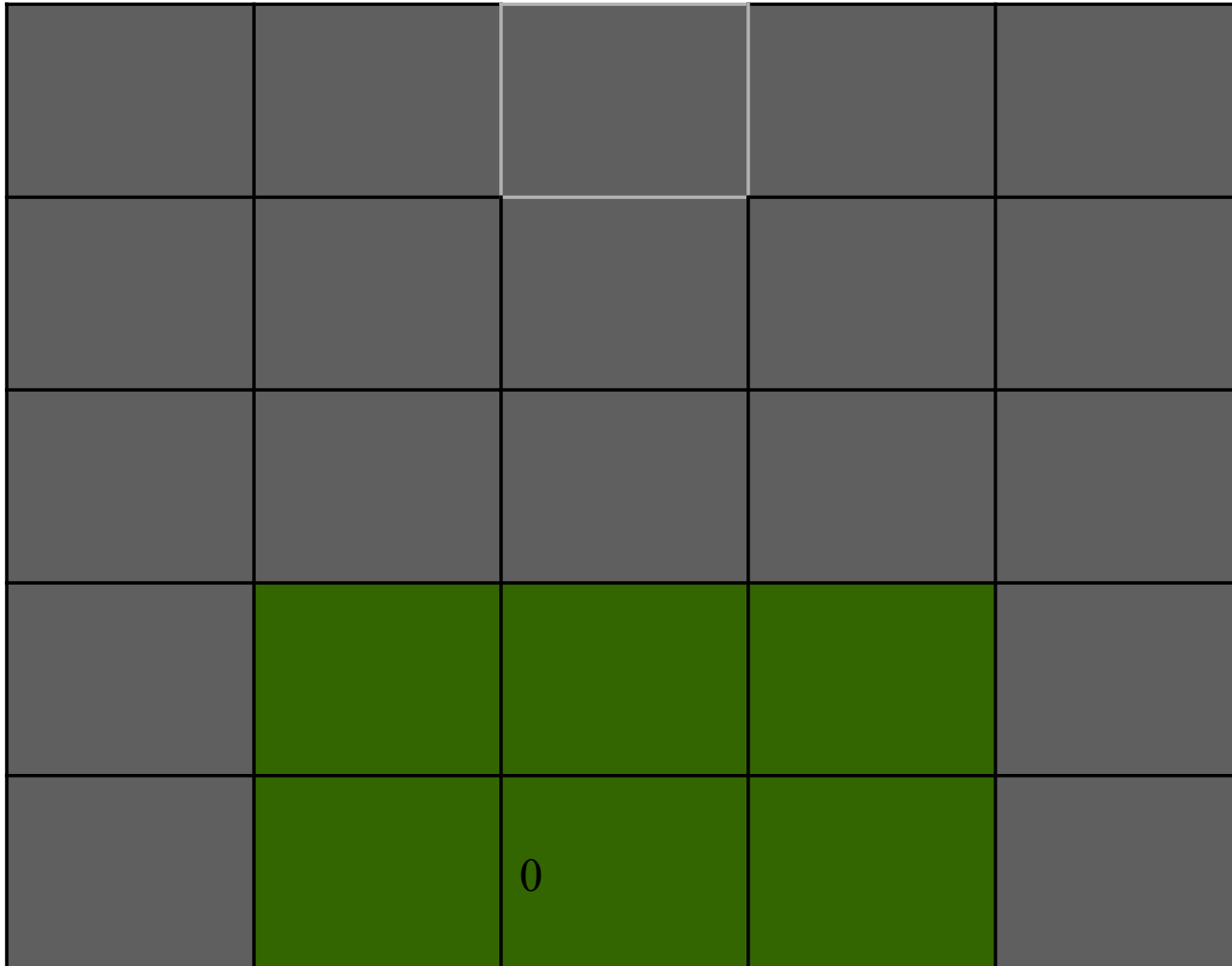


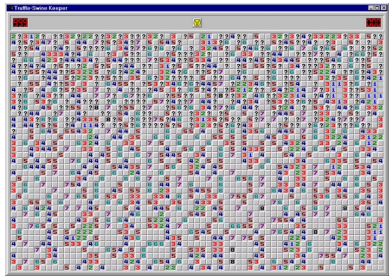
Mine Sweeper



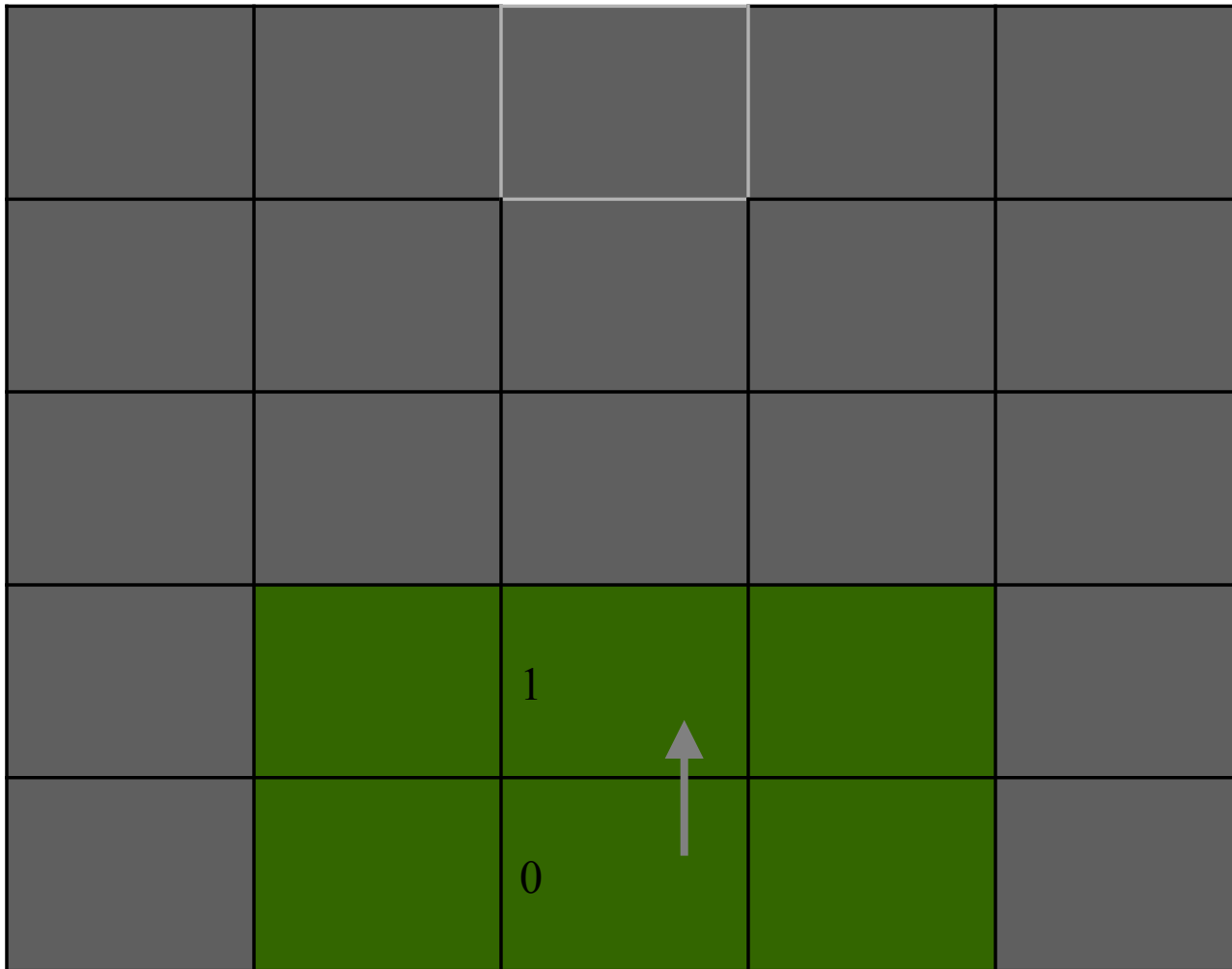


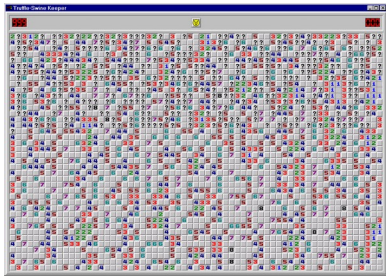
Mine Sweeper



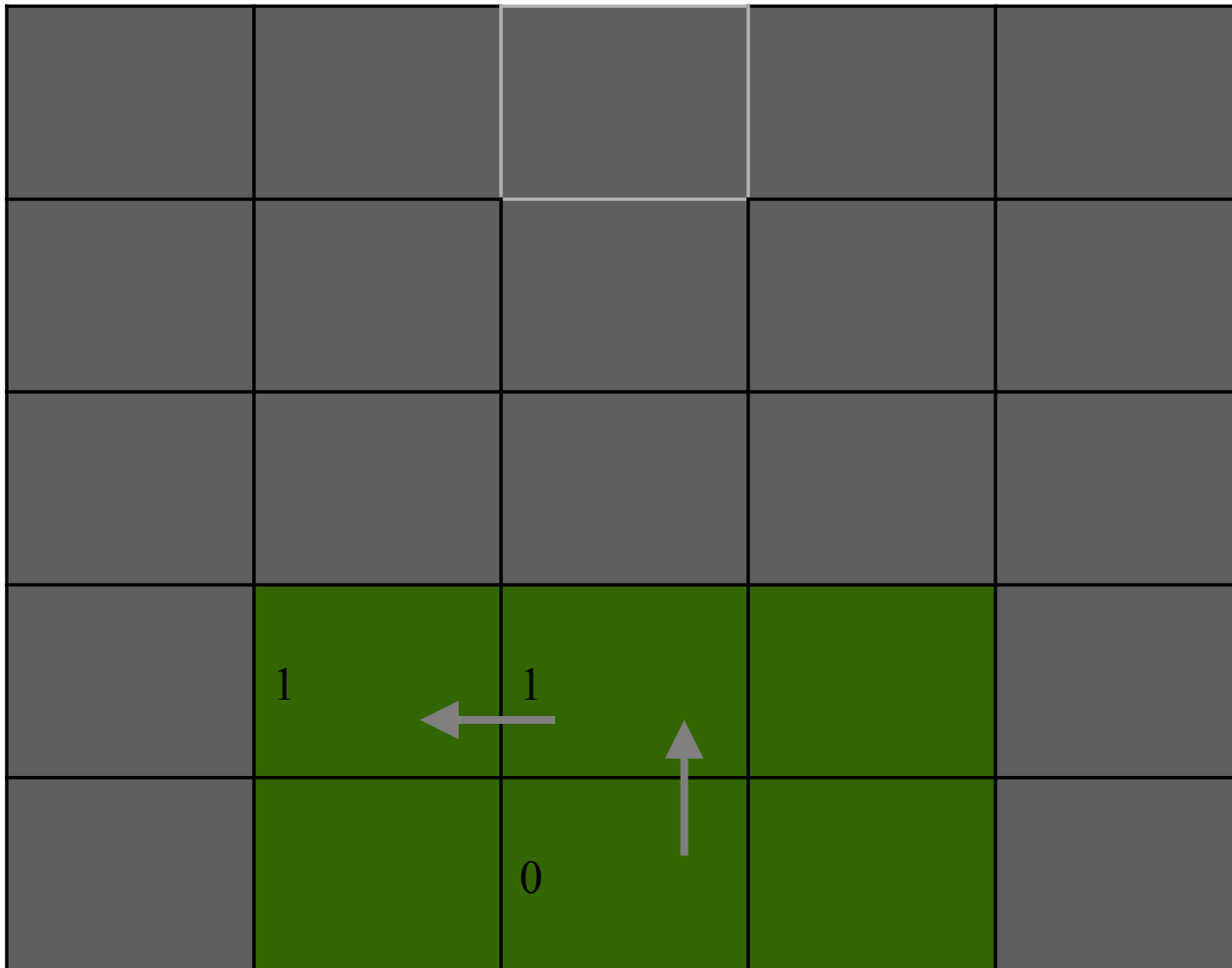


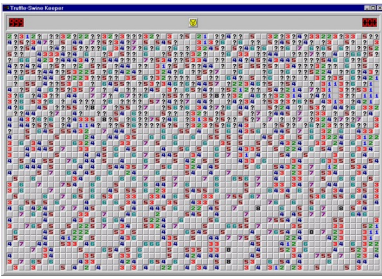
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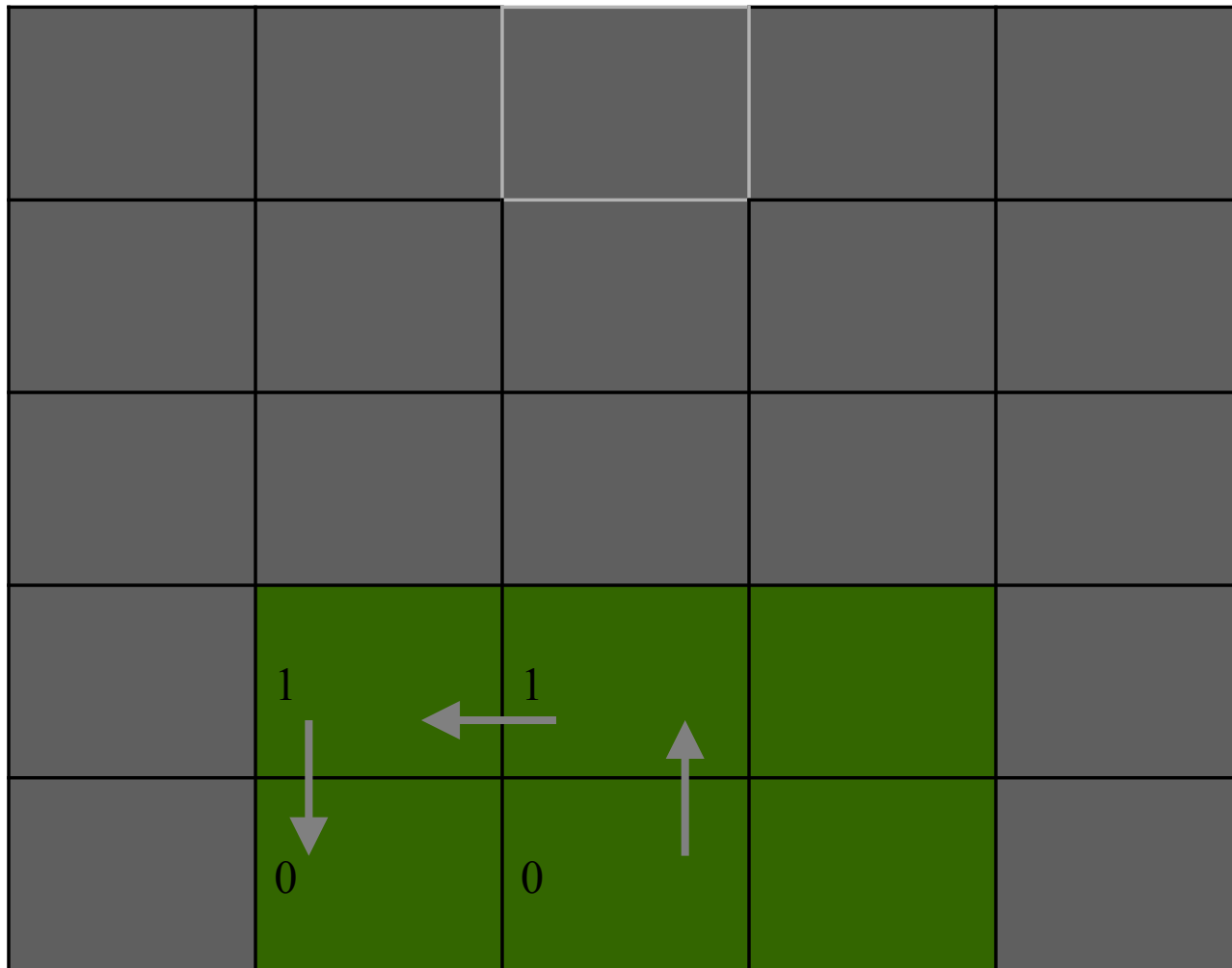


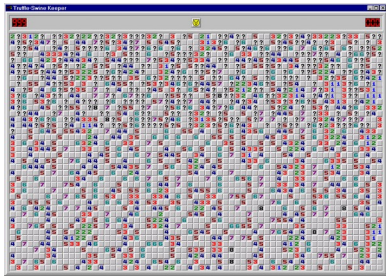
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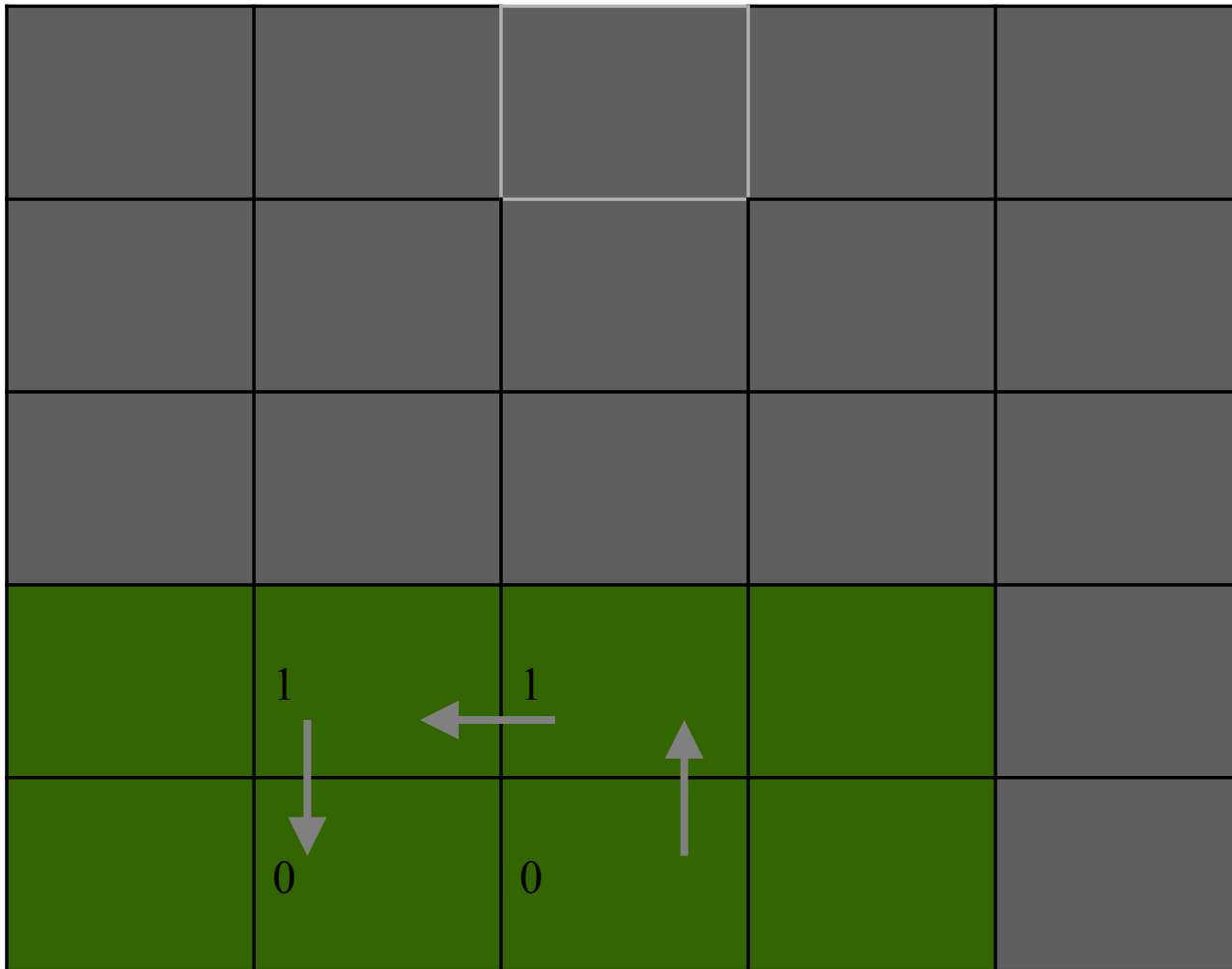


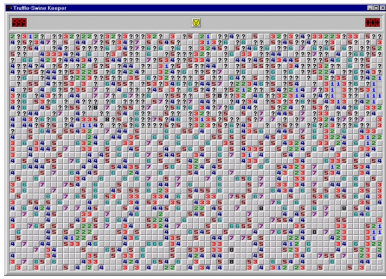
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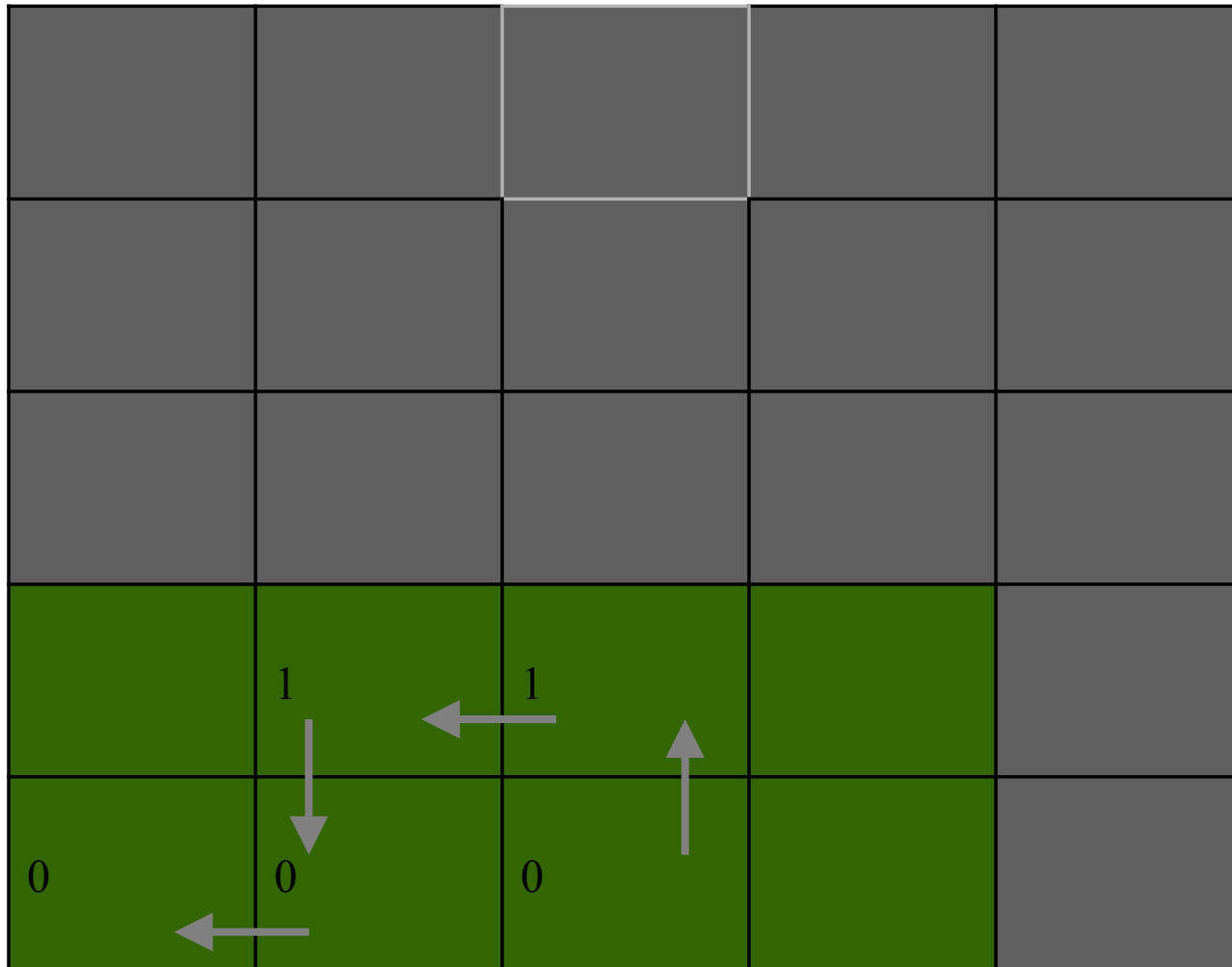


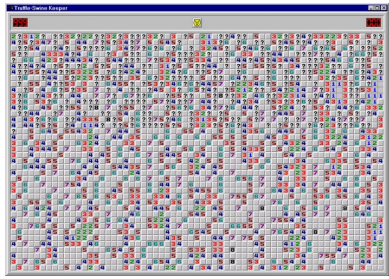
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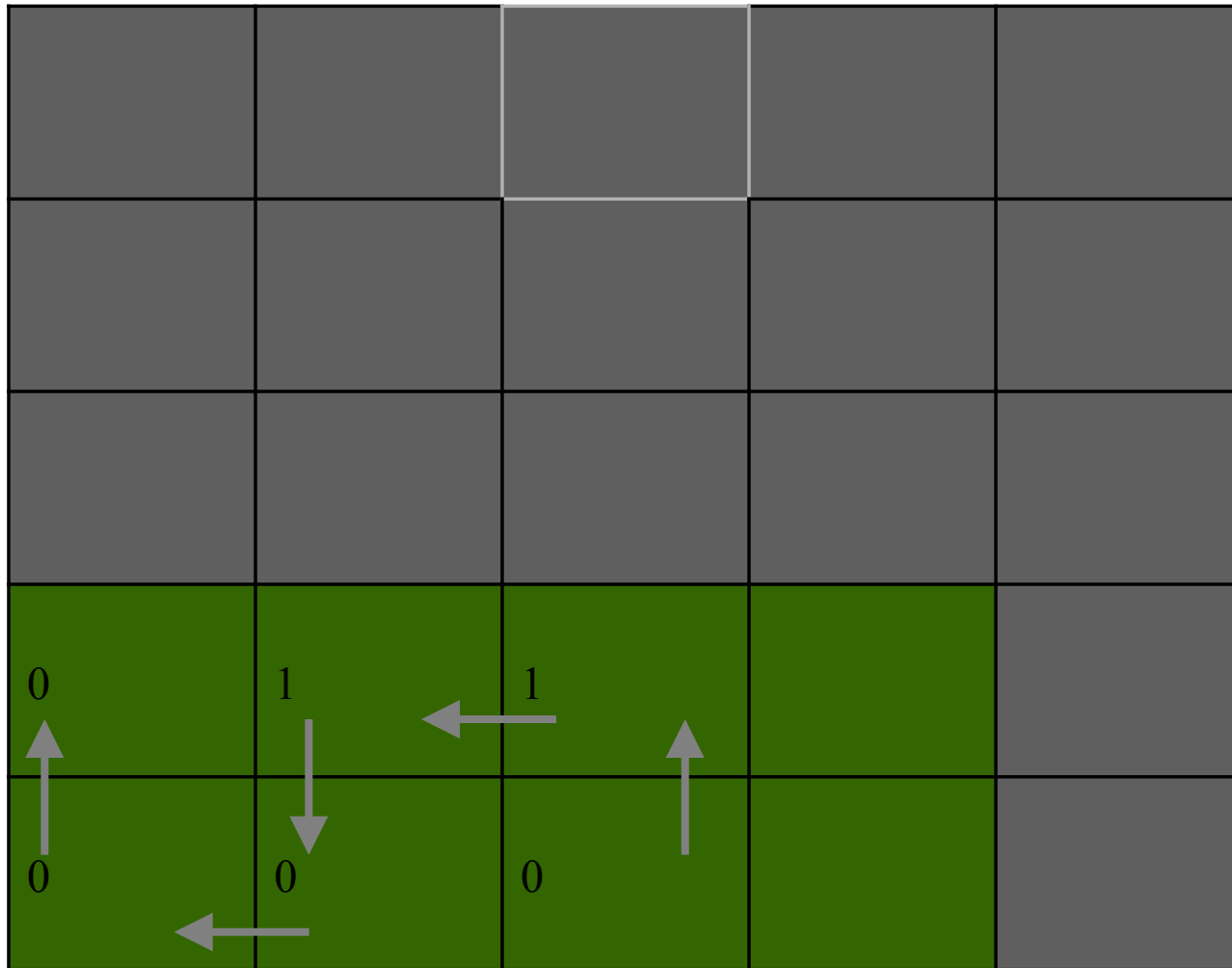


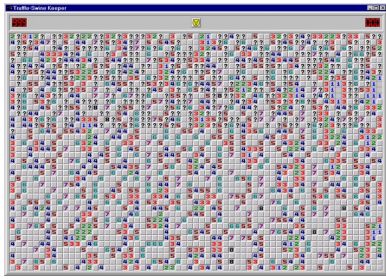
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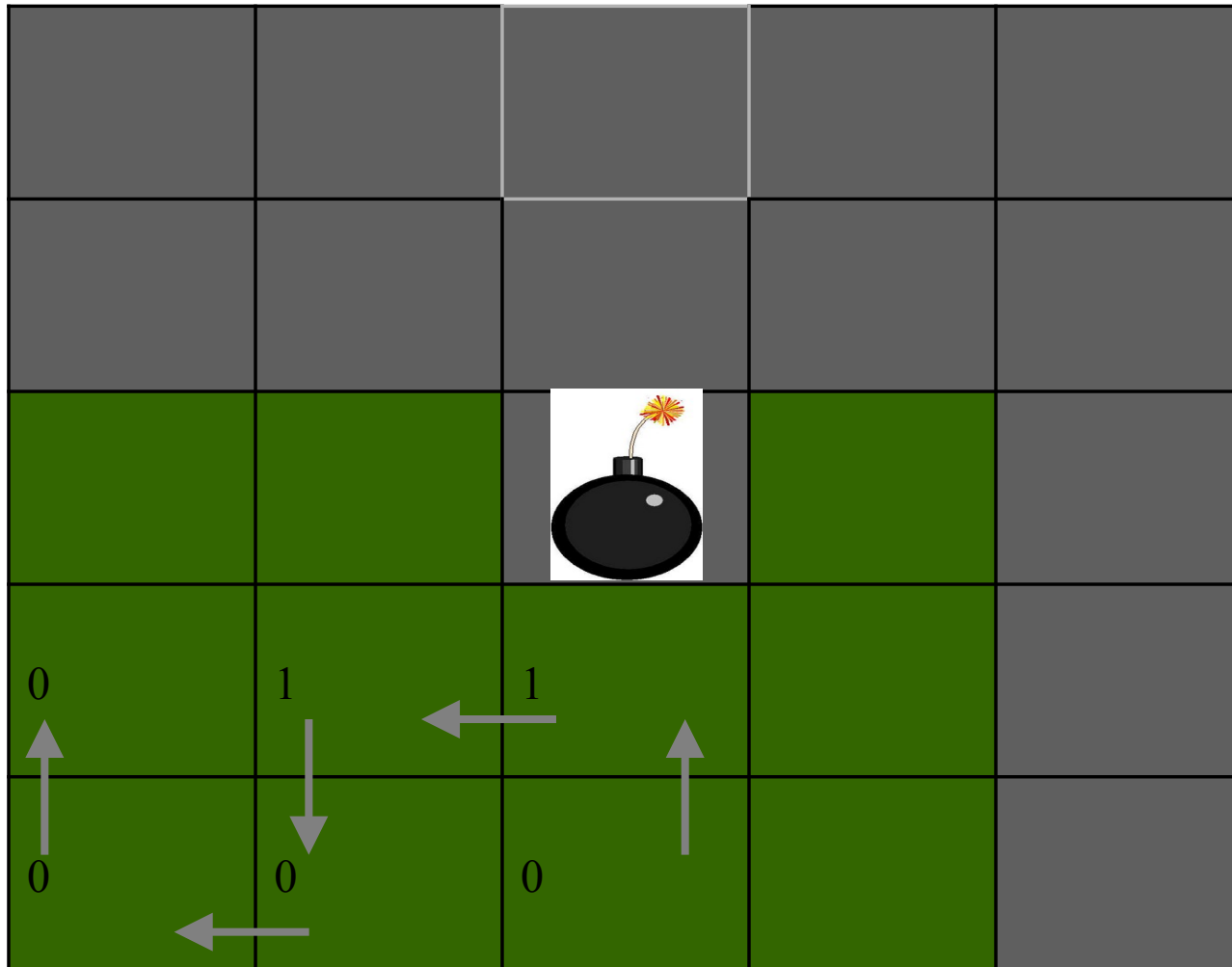


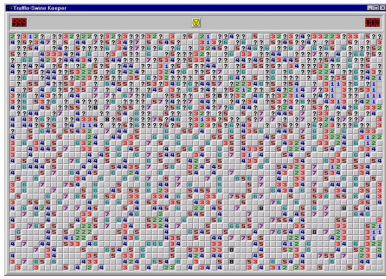
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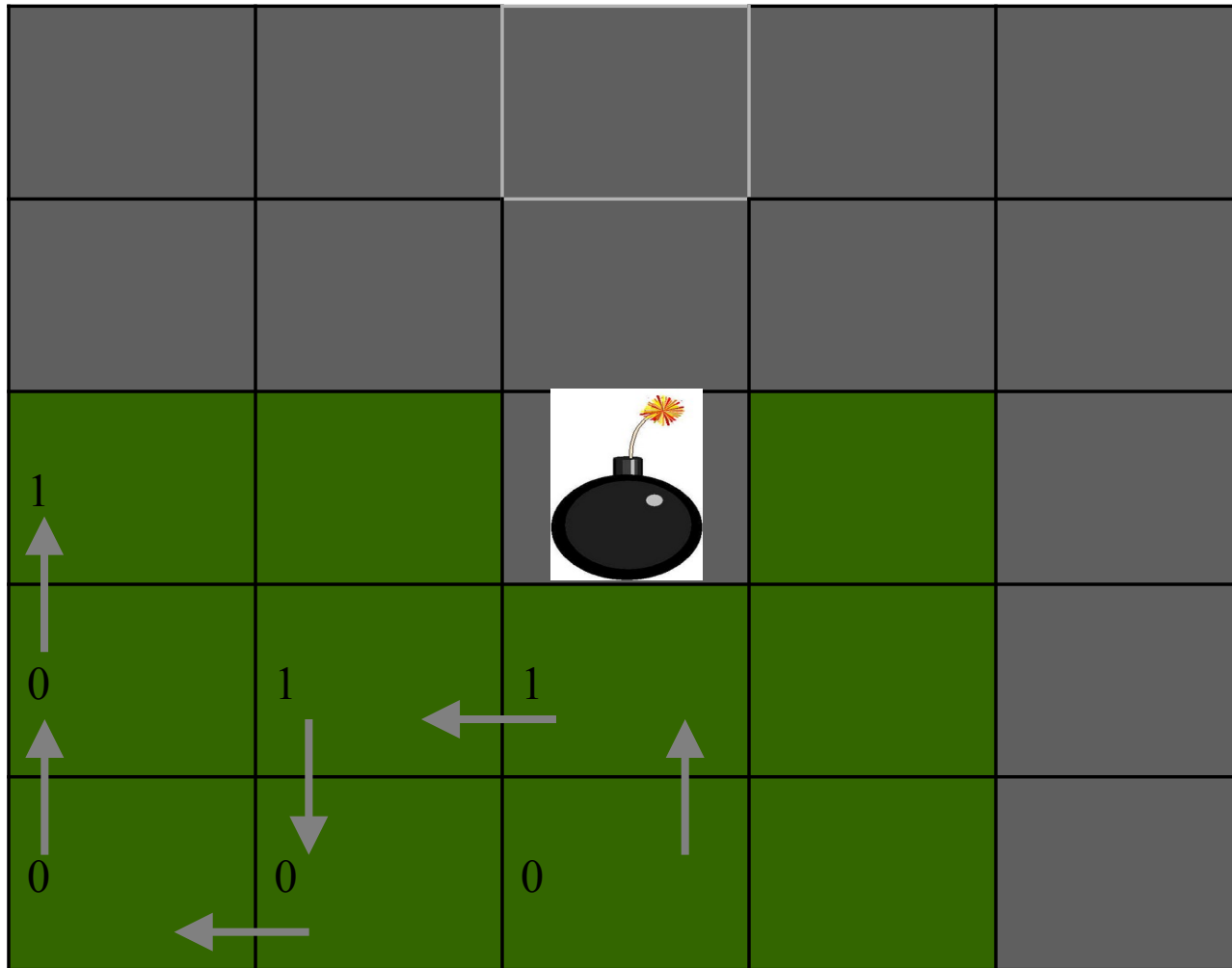


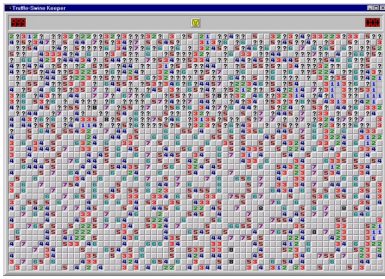
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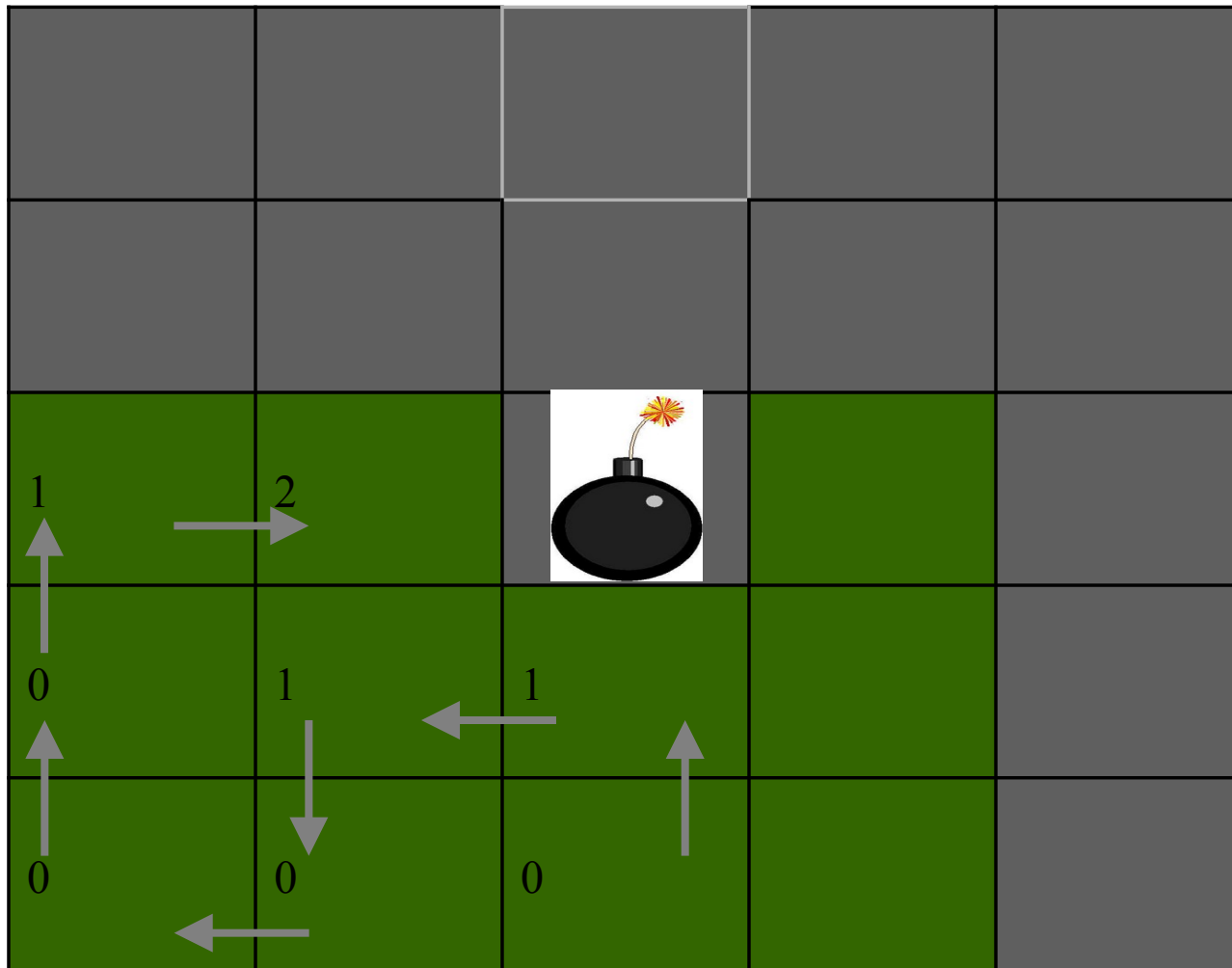


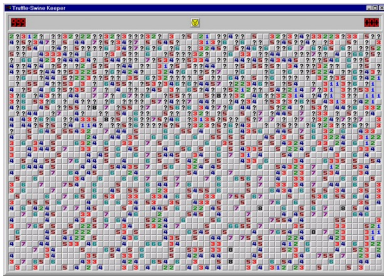
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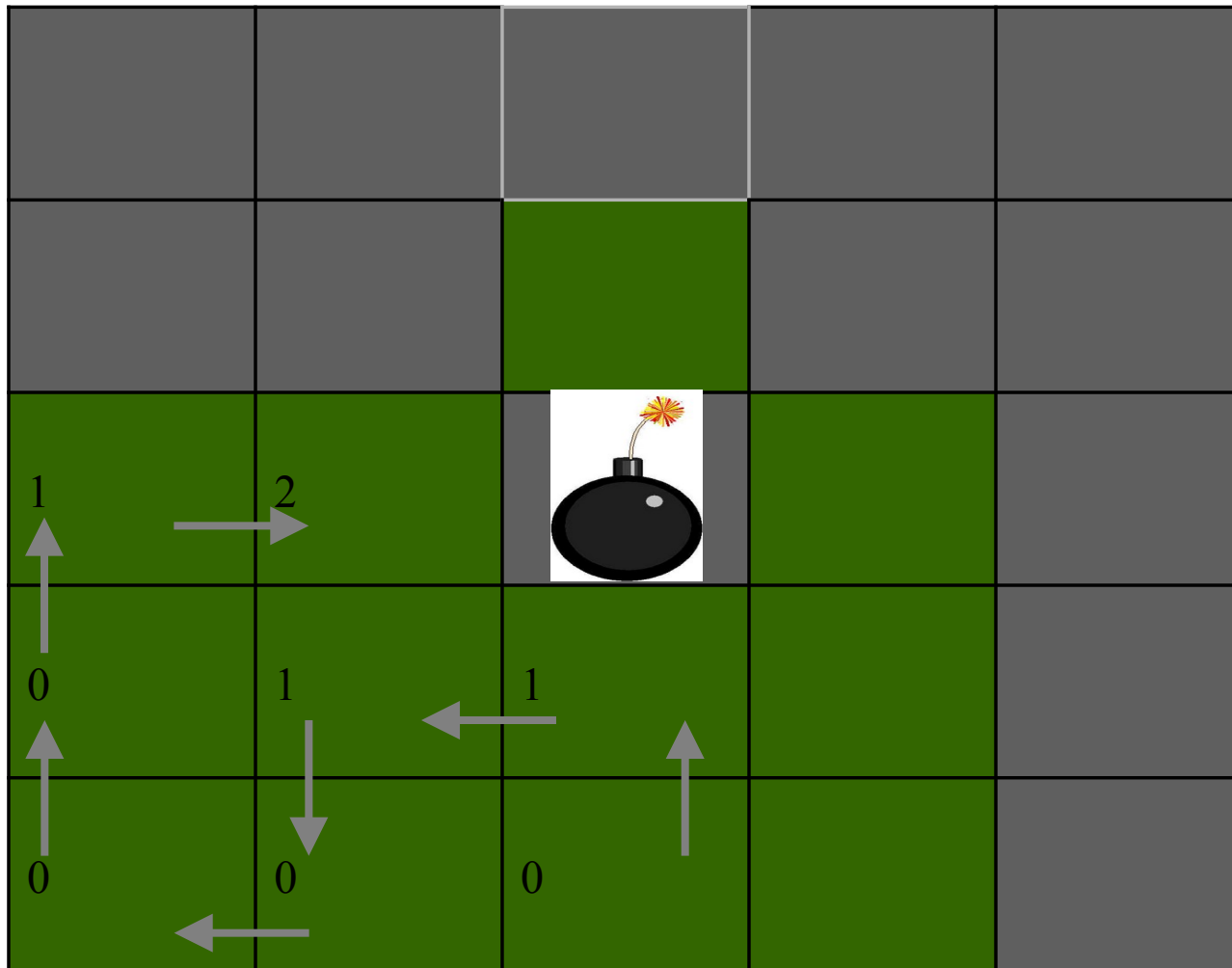


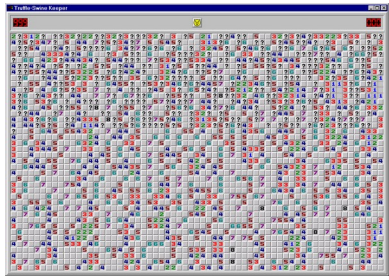
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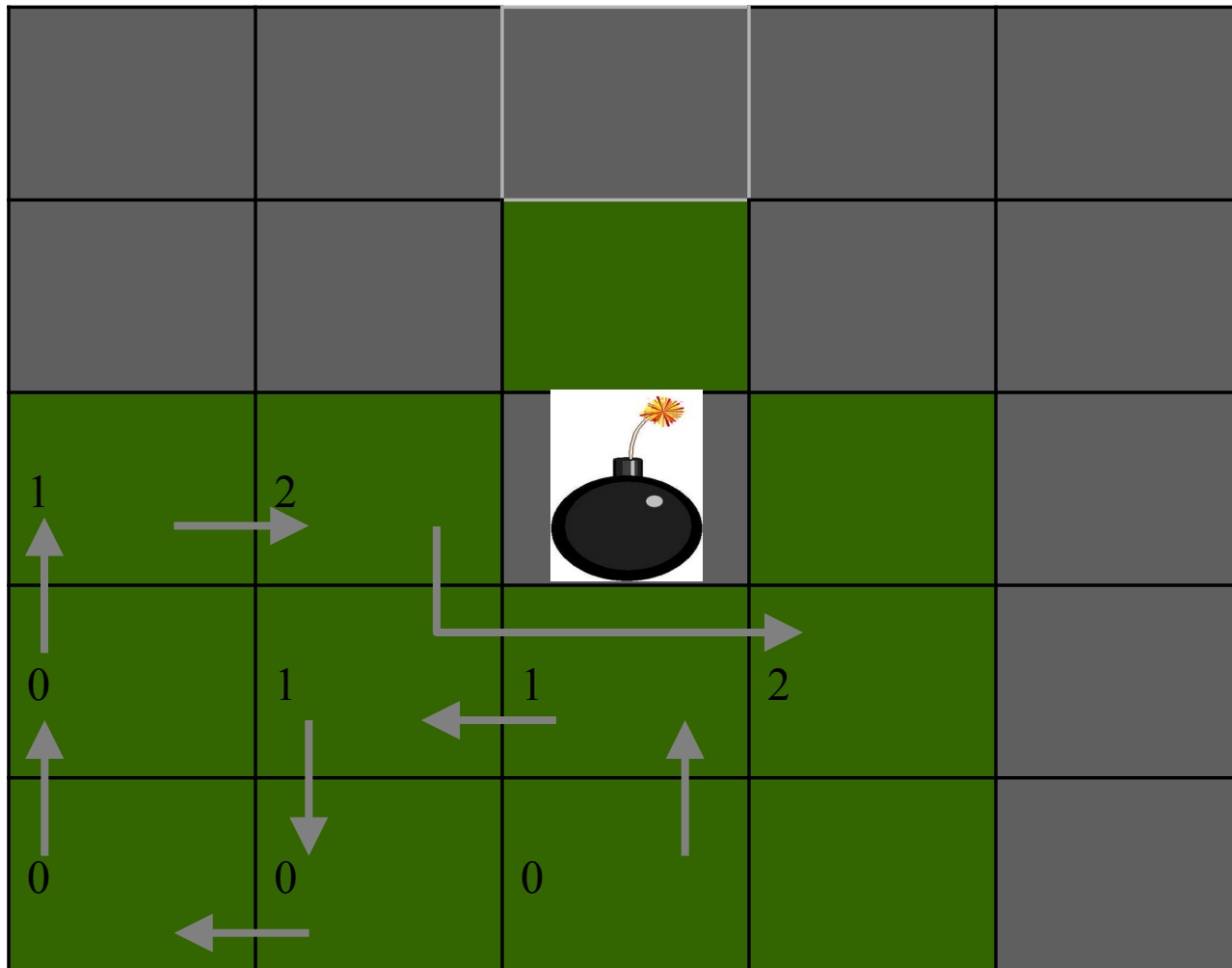


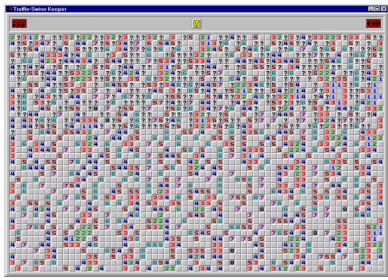
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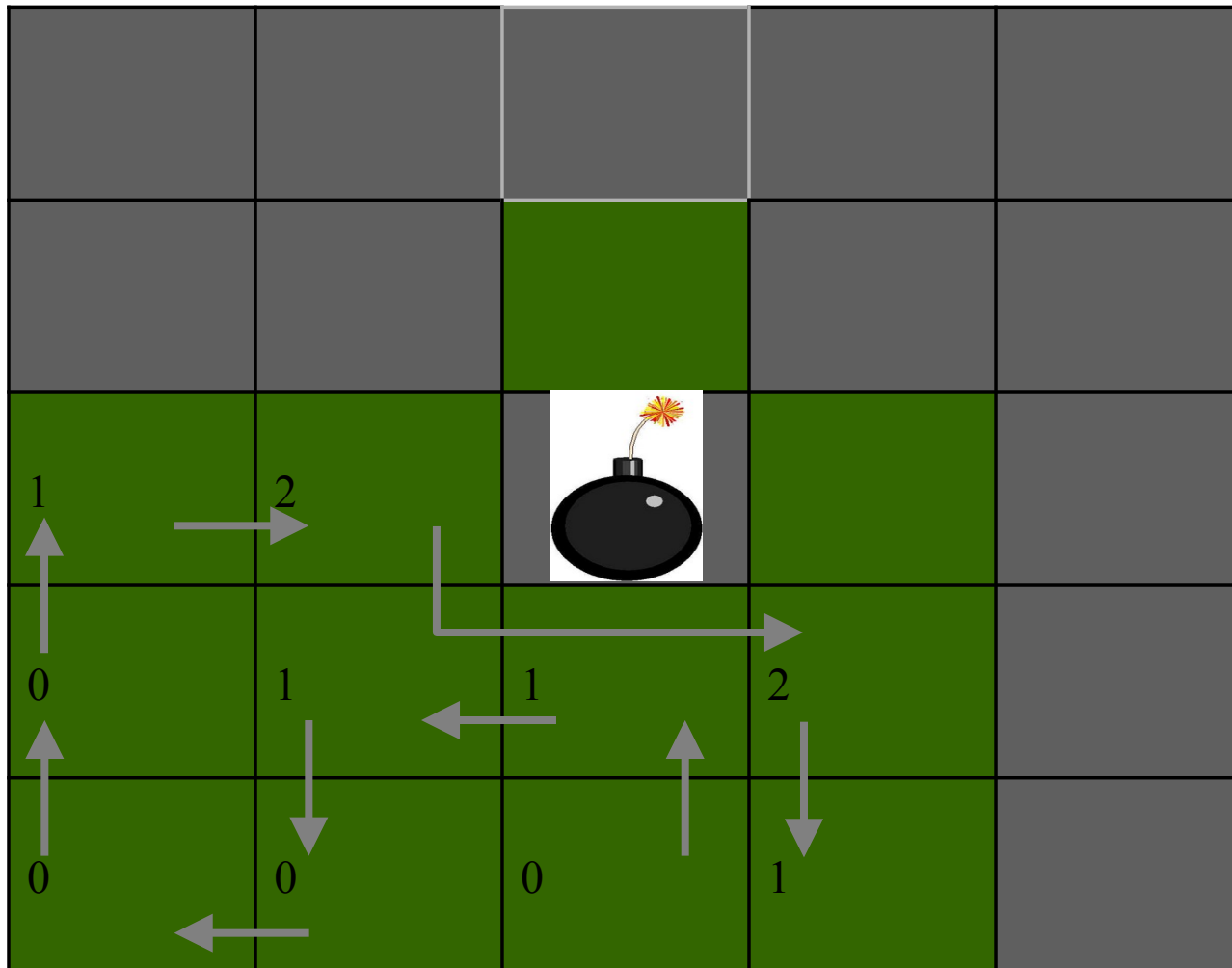


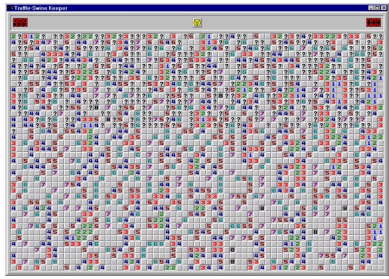
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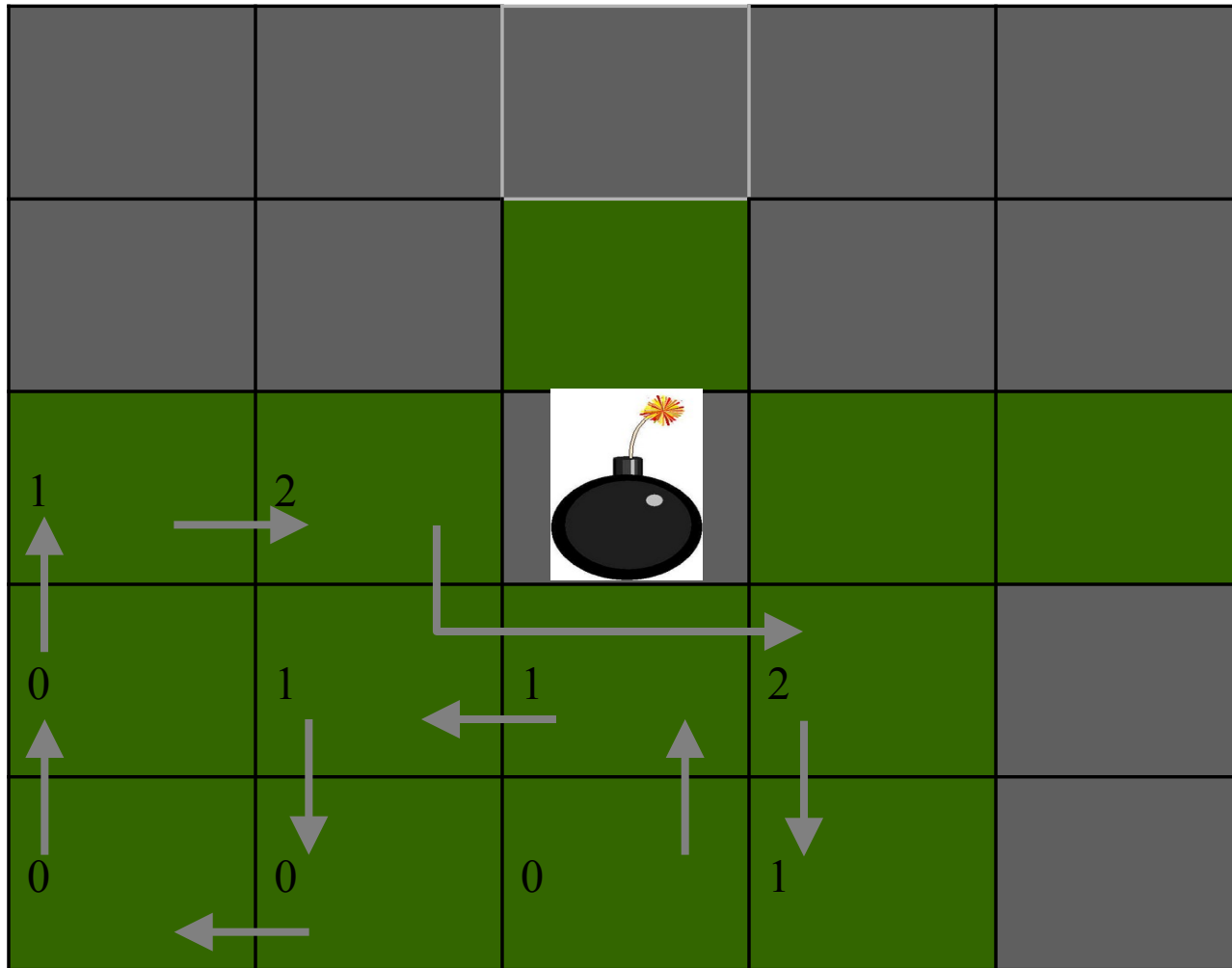


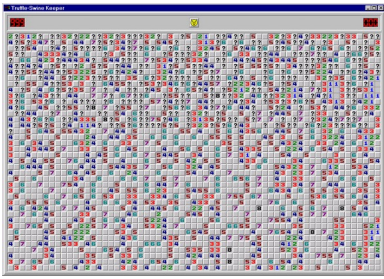
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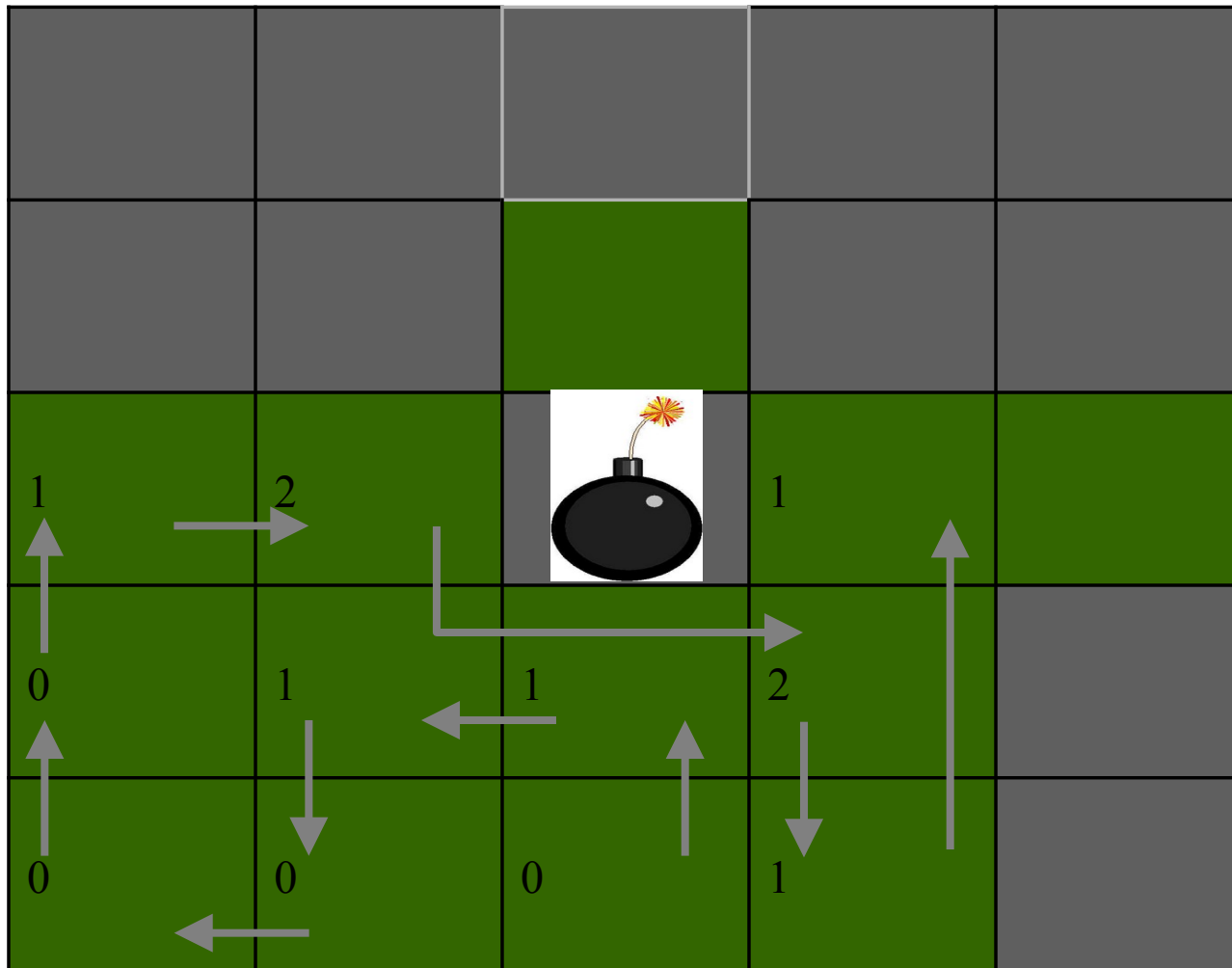


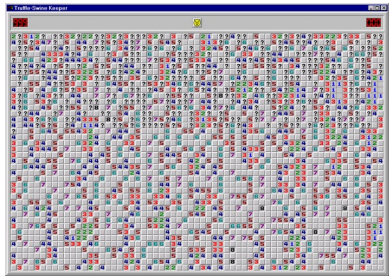
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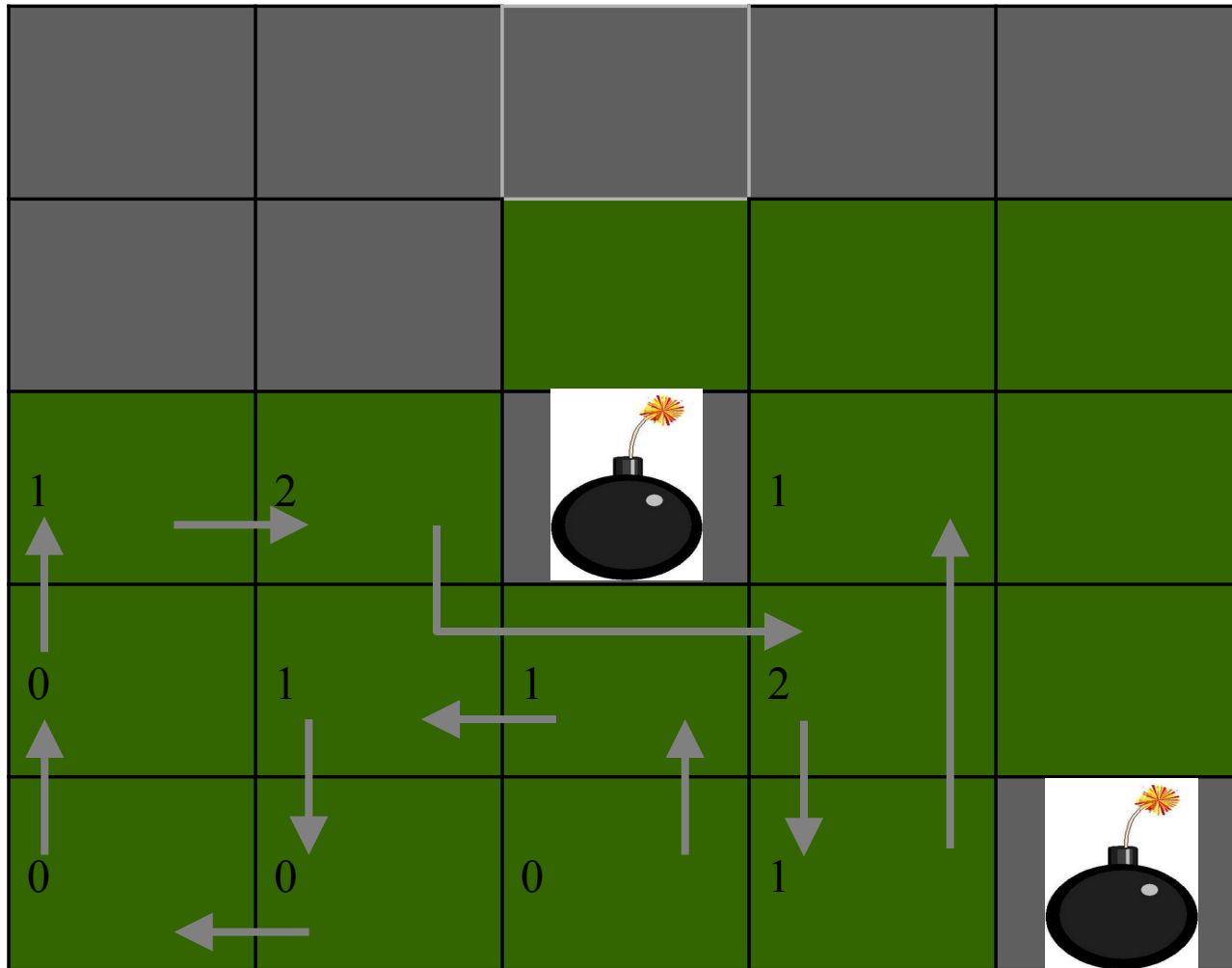


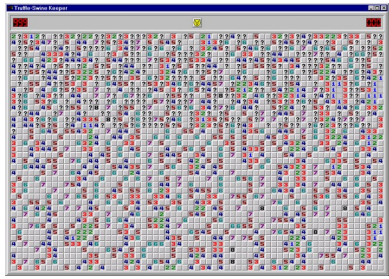
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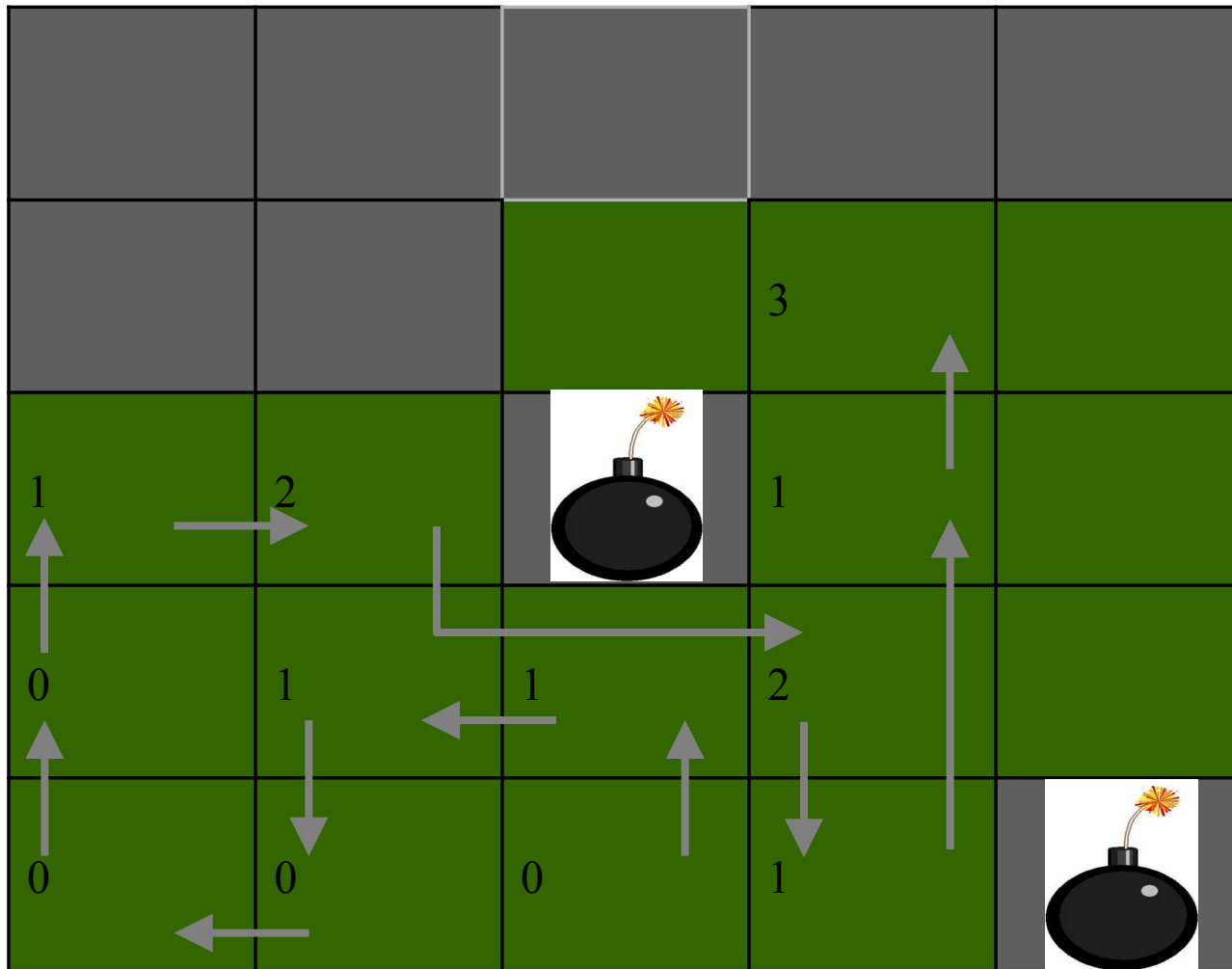


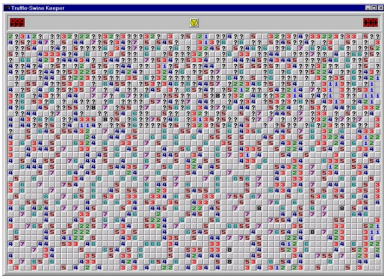
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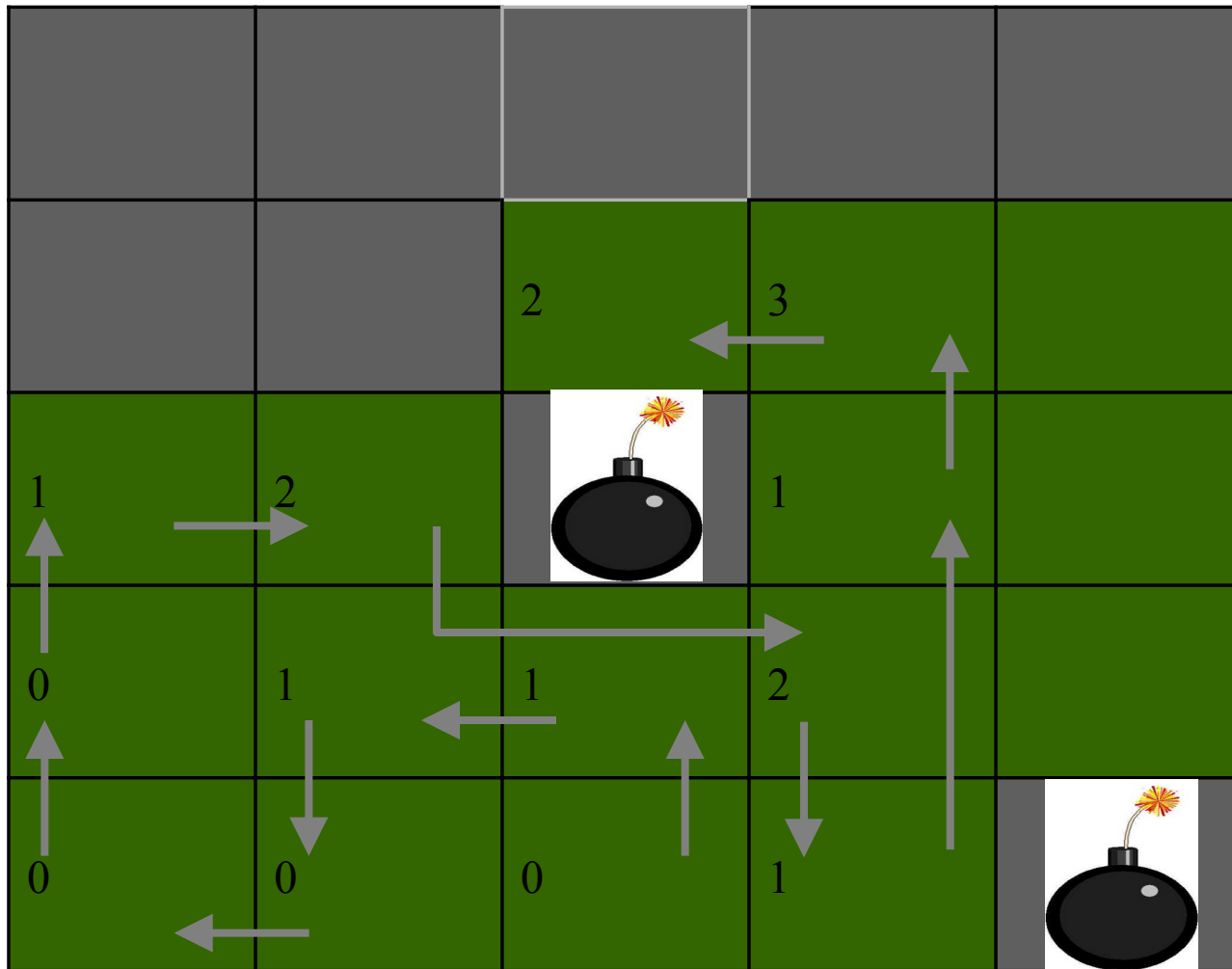


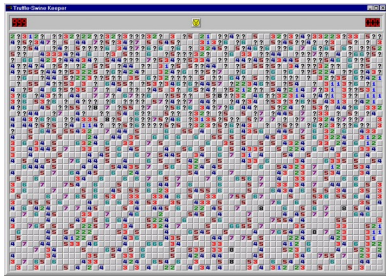
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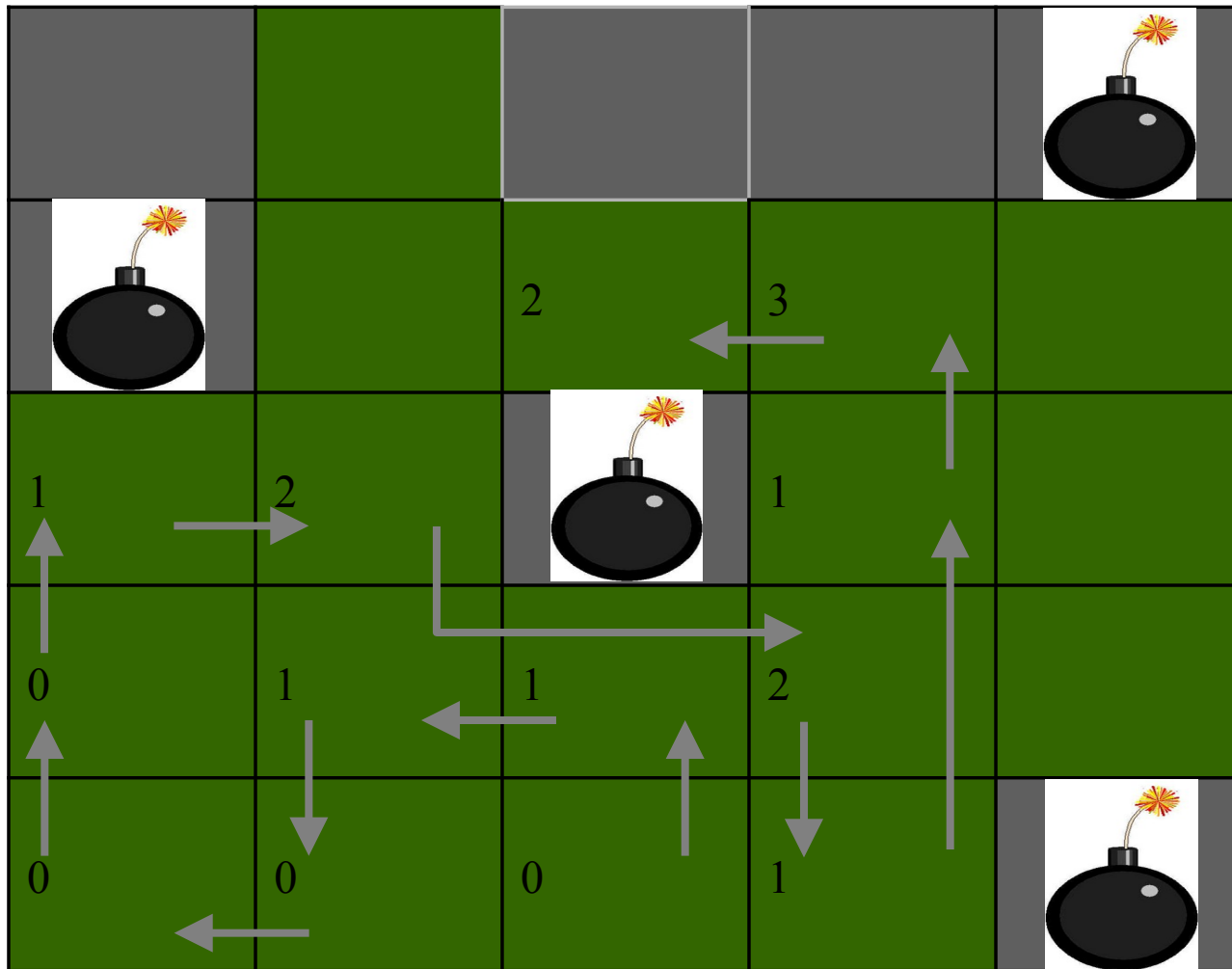


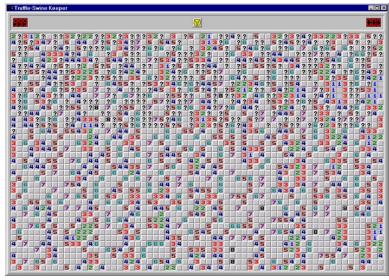
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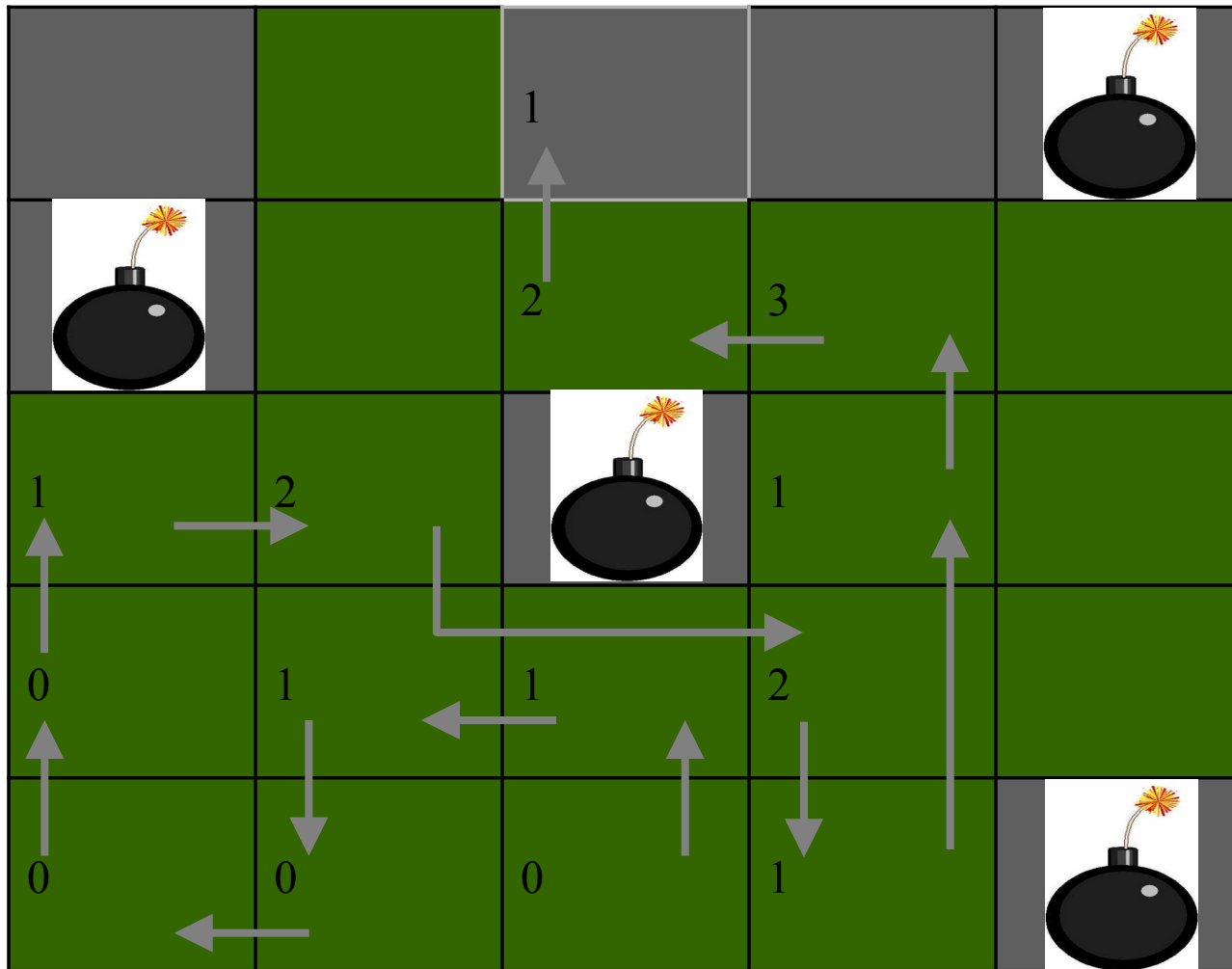


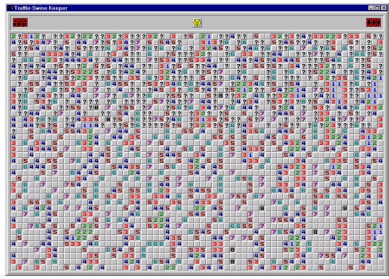
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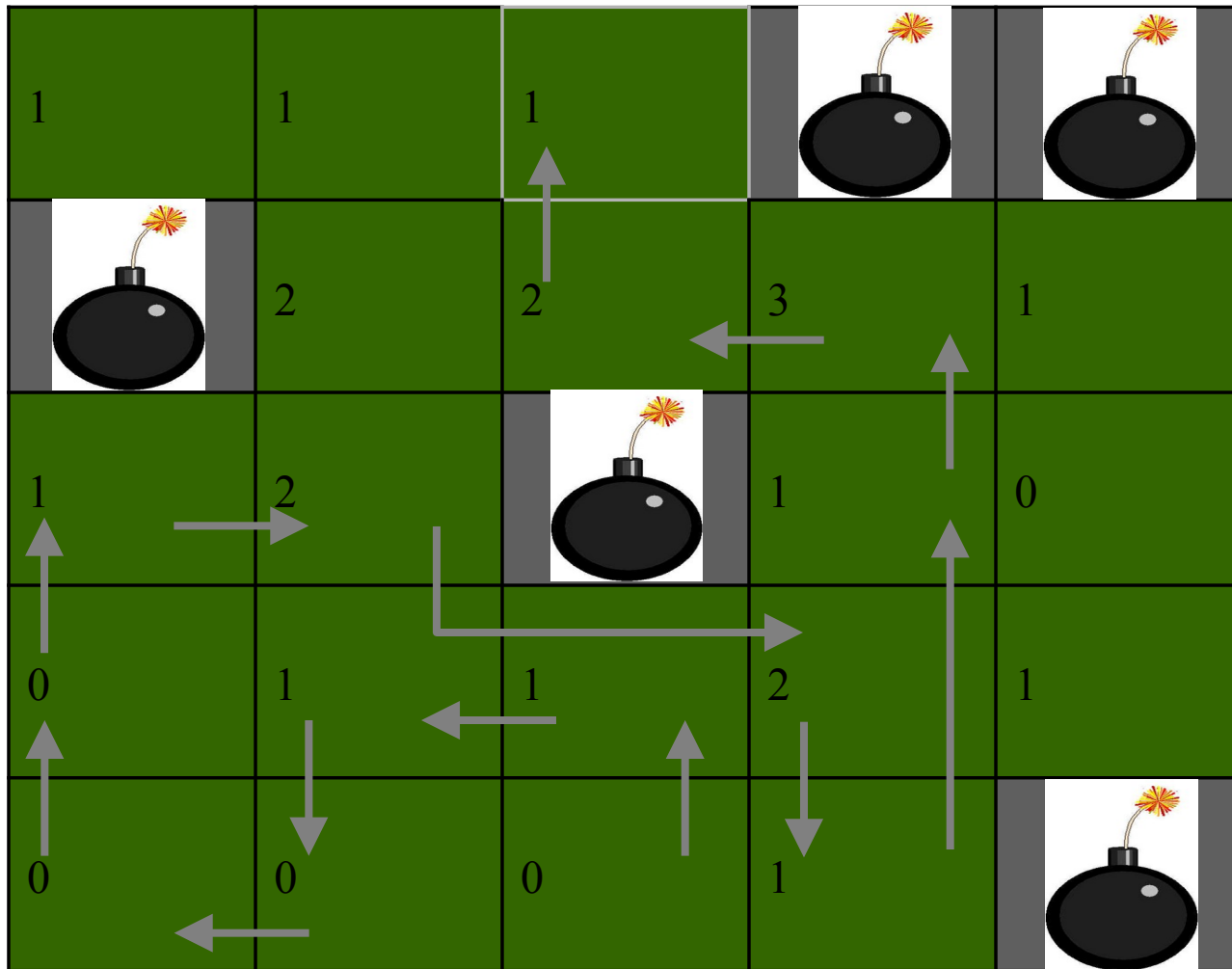


Mine Sweeper





Mine Sweeper



Knowledge Base

Once we have walked on the board for a while, how can we express the knowledge that we have acquired?

The initial cell, for instance, has no bomb. We could express this type of knowledge with 0/1 variables that tell us, for each cell, whether there is a bomb or not: $B_{13} = 0$.

When we see that our device can detect no mines in the proximity of the start field, we can express that knowledge as $\neg B_{12} \wedge \neg B_{14} \wedge \neg B_{22} \wedge \neg B_{23} \wedge \neg B_{24}$

Logic: Outline

1. General principles of logic – main vehicle for representing knowledge
2. Wumpus World - a toy world similar to mine sweeper → it illustrates how a knowledge based agent operates
3. Propositional logic
4. Predicate logic
5. Proof Methods
 1. Inference Rules (including induction)
 2. Model Checking
 - Algorithmic approaches
6. Satisfiability as an Encoding language
7. First Order Logic
8. Inference in First Order Logic

Logic in general

Logics are formal languages for formalizing reasoning, in particular for representing information such that conclusions can be drawn

A **logic** involves:

- A **language** with a **syntax** for specifying what is a legal expression in the language; syntax defines well formed sentences in the language
- Semantics** for associating elements of the language with elements of some subject matter. Semantics defines the "meaning" of sentences (link to the world); i.e., semantics defines the truth of a sentence with respect to each possible world
- Inference rules** for manipulating sentences in the language

Original motivation: Early Greeks, settle arguments based on purely rigorous (symbolic/syntactic) reasoning starting from a given set of premises.

Example of a formal language: Arithmetic

E.g., the language of arithmetic

$-x+2 \geq y$ is a sentence; ($x^2+y > \{\}$ is not a sentence)

$-x+2 \geq y$ is true iff the number $x+2$ is no less than the number y

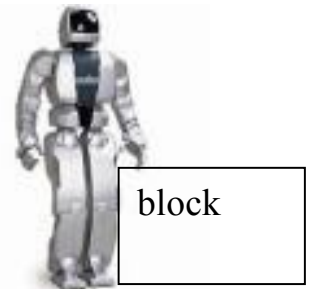
$-x+2 \geq y$ is true in a world where $x = 7, y = 1$

$-x+2 \geq y$ is false in a world where $x = 0, y = 6$

Simple Robot Domain

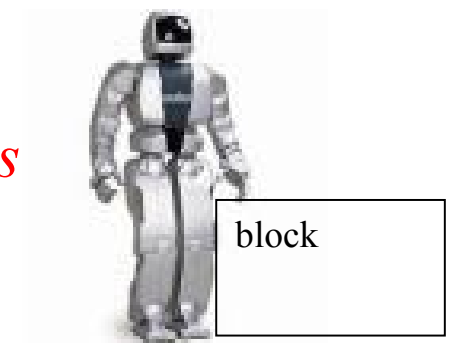
Consider a robot that is able to lift a block, if that block is liftable (i.e., not too heavy), and if the robot's battery power is adequate. If both of these conditions are satisfied, then when the robot tries to lift a block it is holding, its arm moves.

- Feature 1: BatIsOk (True or False)
- Feature 2: BlockLiftable (True or False)
- Feature 3: RobotMoves (True or False)



Simple Robot Domain

We need a *language* to express the *features/properties/assertions* and *constraints among them*; also *inference mechanisms*, *i.e.*, principled ways of performing reasoning.



Example logical statement about the robot:

(BatIsOk **and** BlockLiftable) **implies** RobotMoves

Logical Reasoning: Entailment

We can think of a knowledge base as a *statement* and we talk about a *knowledge base entailing a sentence*.

Entailment means that one thing **follows from** another:

$$KB \models \alpha$$

A Knowledge base *KB* **entails** sentence α **iff** α **is true in all worlds where *KB* is true**

–E.g., the KB containing “Giants won” and “Reds won” entails “Either the Giants won or the Reds Won” Why?

–E.g., $x+y = 4$ entails $4 = x+y$

–Entailment is a relationship between sentences (i.e., **syntax**) that is based on **semantics**

Models

Logicians typically think in terms formally **structured worlds with respect to which truth can be evaluated;**

Example:

– $x + y \geq 7$, is true in all the worlds in which $x \geq 7 - y$, assuming that we are dealing with real numbers, in particular $x = 7$ and $y = 0$ or $x = 8$ and $y = 1$, etc ;

– Basically, each world corresponds to a different assignment to the variables satisfying the constraints; note each assignment determines the truth or falsehood of the arithmetic sentence.

We say **m is a model of a sentence α if α is true in m**

$M(\alpha)$ is the set of all models of **α** (i.e., models that assign true to α).

Wumpus World

Performance measure

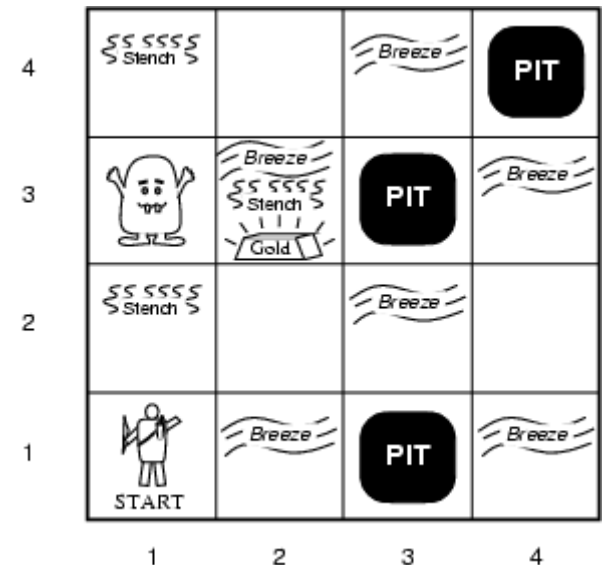
- gold +1000,
- death -1000
(falling into a pit or being eaten by the wumpus)
- 1 per step, -10 for using the arrow

Environment

- Squares adjacent to wumpus are smelly
- Squares adjacent to pit are breezy
- Glitter iff gold is in the same square
- Shooting kills wumpus if you are facing it
- Shooting uses up the only arrow
- Grabbing picks up gold if in same square
- Releasing drops the gold in same square

Sensors: Stench, Breeze, Glitter, Bump, Scream

Actuators: Left turn, Right turn, Forward, Grab, Release, Shoot



Models

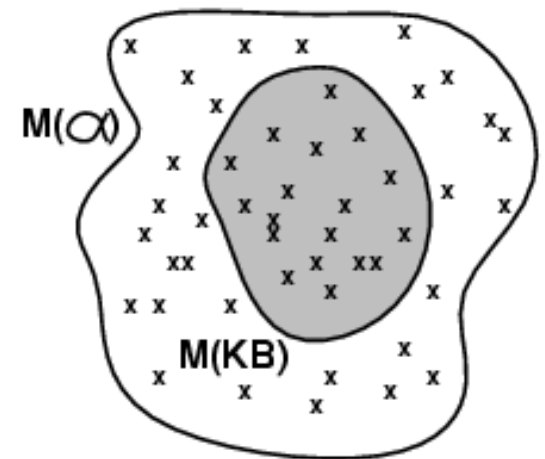
$$KB \models \alpha \text{ iff } M(KB) \subseteq M(\alpha)$$

–E.g. KB = Giants won and Reds Won

– α = Giants won or Reds Won

–Other ways of talking about entailment:

$$KB \models \alpha$$



2 - Wumpus World

Wumpus world characterization

Fully Observable No – only **local** perception

Deterministic Yes – outcomes exactly specified

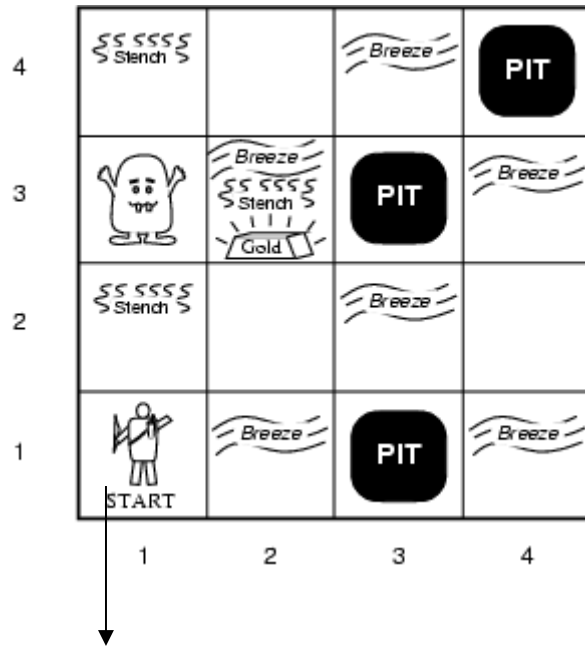
Episodic No – sequential at the level of actions

Static Yes – Wumpus and Pits do not move

Discrete Yes

Single-agent? Yes – Wumpus is essentially a natural feature

Exploring a wumpus world

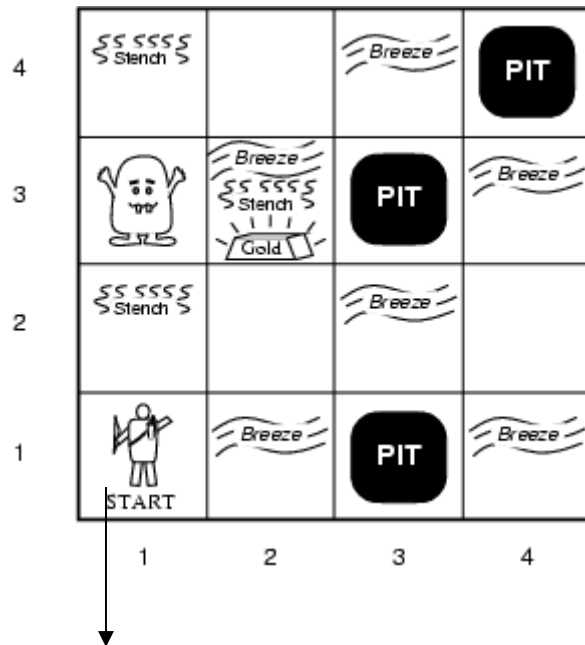


The knowledge base of the agent consists of the rules of the Wumpus world plus the percept nothing in [1,1]

None, none, none, none, none

Stench, Breeze, Glitter, Bump, Scream

Exploring a wumpus world



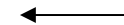
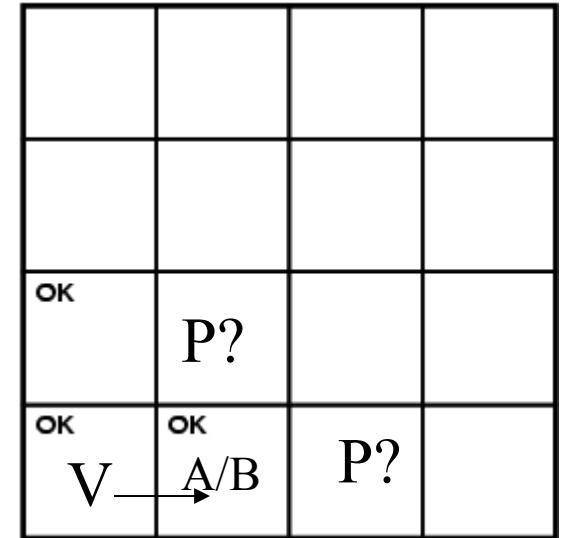
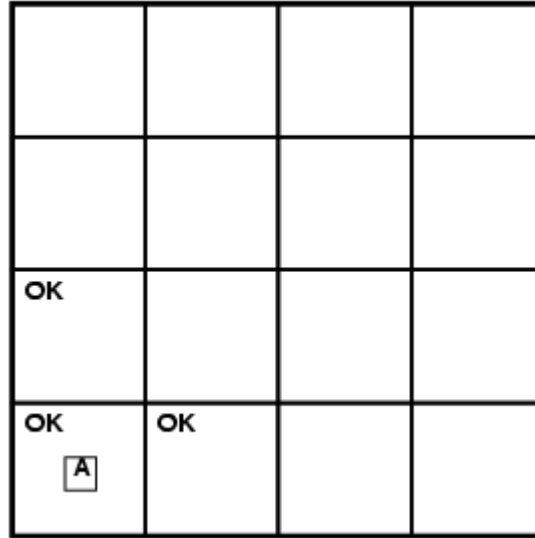
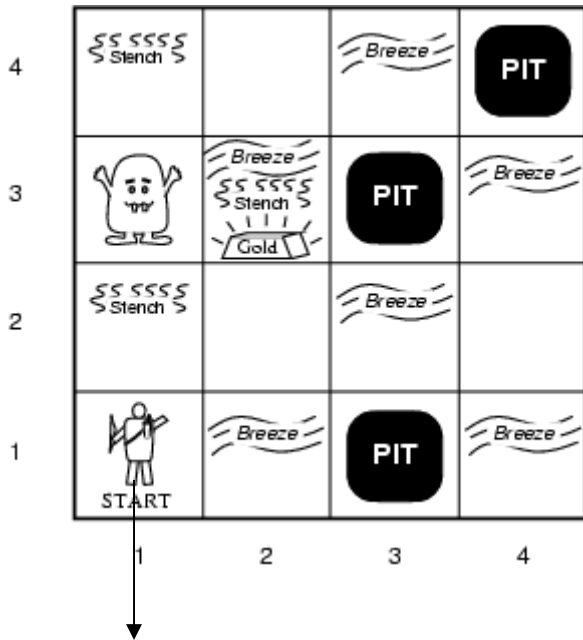
None, none, none, none, none

Stench, Breeze, Glitter, Bump, Scream

OK			
OK A	OK		

The knowledge base of the agent consists of the rules of the Wumpus world plus the percept nothing in [1,1]; by inference, the agent's knowledge base also has the information that [2,1] and [1,2] are okay.

Exploring a wumpus world



None, breeze, none, none, none

None, none, none, none, none

Stench, Breeze, Glitter, Bump, Scream

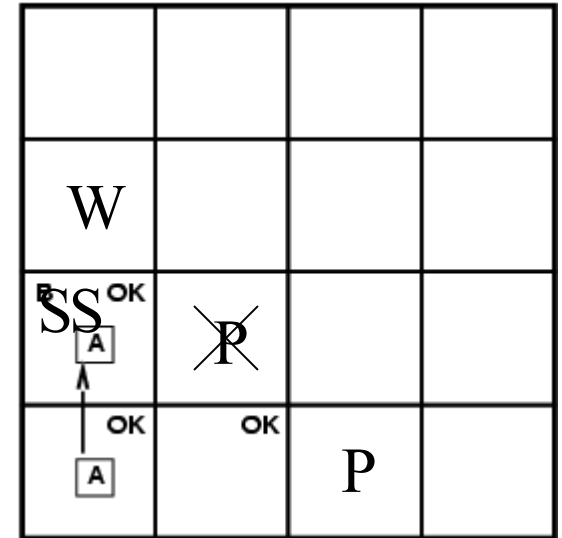
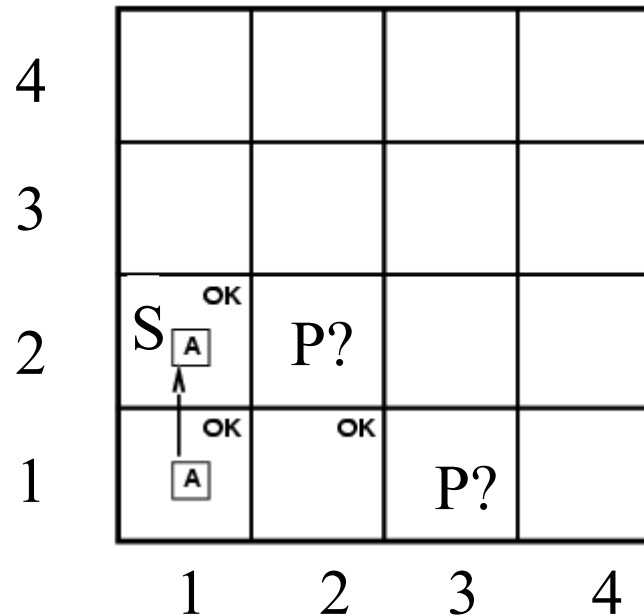
A – Agent

V – visited

B - Breeze

Pit in (2,2) or (3,1)

Exploring a wumpus world



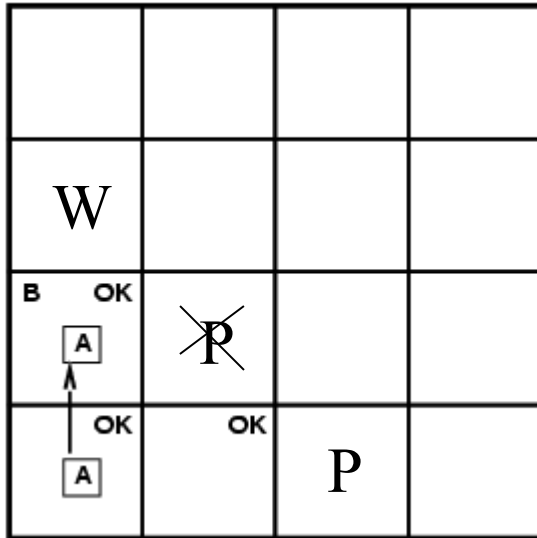
S = (Stench, none, none, none, none)

S → Wumpus nearby

Wumpus cannot be in (1,1) or in (2,2) (Why?) → Wumpus in (1,3)

Not breeze in (1,2) → no pit in (2,2); but we know there is a pit in (2,2) or (3,1) → pit in (3,1)

Exploring a wumpus world



Difficult inference, because it combines knowledge gained at different times in different places; this **inference is beyond the abilities of most animals.**

In each case where the agent draws a conclusion from the available information, that conclusion is **guaranteed to be correct** if the initial information is correct - fundamental property of logical reasoning!

Entailment in the wumpus world

Knowledge Base in the Wumpus World →

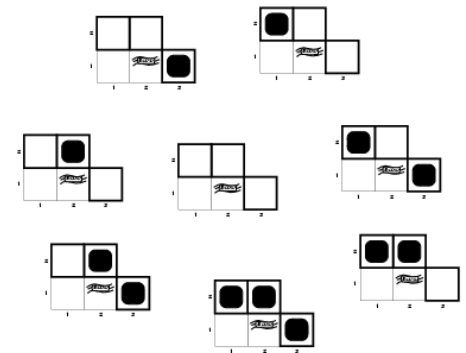
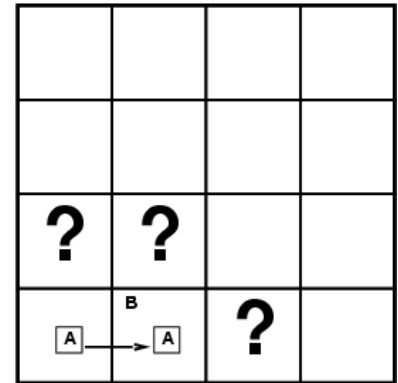
Rules of the wumpus world + new percepts

Situation after detecting nothing in [1,1], moving right, breeze in [2,1]

Consider possible models for *KB* with respect to the cells (1,2), (2,2) and (3,1), with respect to the existence or non existence of pits

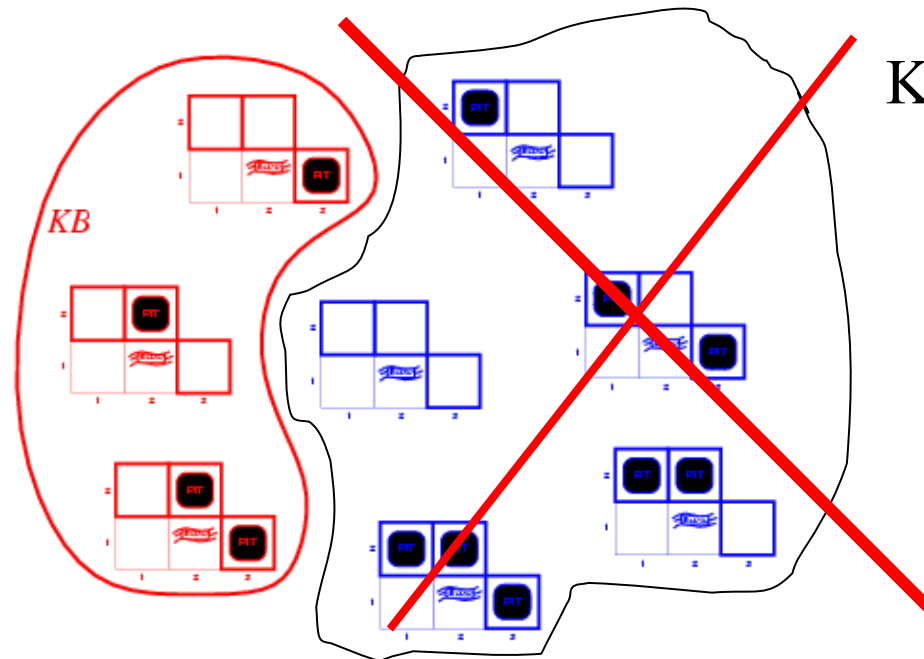
3 Boolean choices ⇒

8 possible interpretations (enumerate all the models)



Wumpus models

IS the KB true in
all of them?

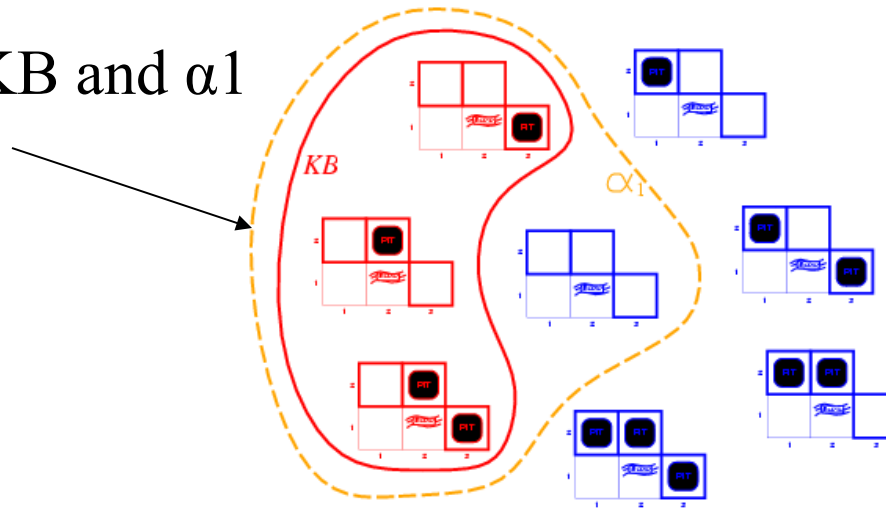


KB false

$KB = \text{wumpus-world rules} + \text{observations}$

Entailment in Wumpus World

Models of the KB and α_1

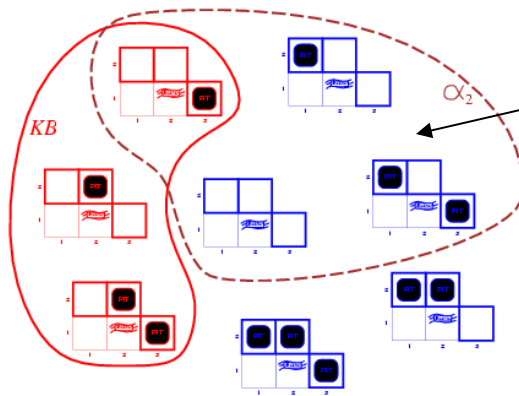


KB = wumpus-world rules + observations

α_1 = "[1,2] has no pit", $KB \models \alpha_1$,

–In every model in which KB is true, α_1 is True (proved by **model checking**)

Wumpus models



Models of the KB and α_2

KB = wumpus-world rules + observations

α_2 = "[2,2] has no pit", this is only True in some of the models for which KB is True, therefore $KB \not\models \alpha_2$
Inference algorithm used to reason about α_1 and $\alpha_2 \rightarrow$

Model Checking

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Inference: Model Checking

Inference by Model checking –

we enumerate all the KB models and check if α_1 and α_2 are True in all the models (which implies that we can only use it when we have a finite number of models).

Inference

$KB \vdash_i \alpha$ we say sentence α can be derived from KB by procedure i

Soundness (or Truth preservation): i is sound if whenever $KB \vdash_i \alpha$, it is also true that $KB \models \alpha$; an unsound procedure can conclude statements that are not true.

Completeness: i is complete if whenever $KB \models \alpha$, it is also true that $KB \vdash_i \alpha$; a complete procedure is able to derive any sentence that is entailed. That is, the procedure will answer any question whose answer follows from what is known by the KB .

Preview: we will define a logic -- **first-order logic**:

- First-order logic is **expressive enough to say almost anything of interest**;
- There exists a **sound and complete inference procedure for first-order logic**, i.e., the procedure will answer any question whose answer follows from what is known by the KB.

As a starting point we will consider a restricted form of first order logic, **propositional logic**.

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